Design Optimization of Solar Parabolic Trough Collector

B. Tech. Phase II Major Project Report submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology (B.Tech.)

In

MECHANICAL ENGINEERING

By

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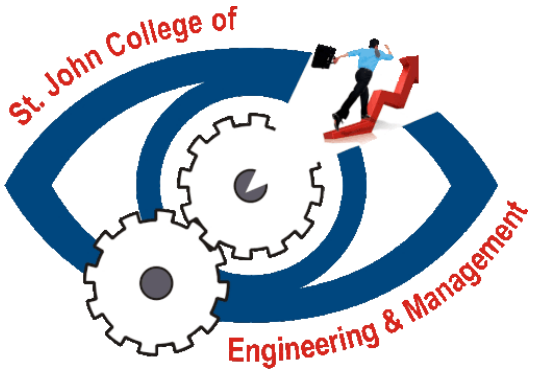
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2026 – 2027

CERTIFICATE

This is to certify that the B.E. project entitled "Design Optimization of Solar Parabolic Trough Collector" is a bonafide work of "Nitin Jagat Singh Verma" (EU1225002), "Hardik Manoj Naik" (EU1225007), Vikram Shrivishnu Mestry" (EU1235002) and "Sachit Ashish Patankar" (EU2225002) submitted to University of Mumbai in partial fulfilment of the requirement for the award of the degree of "Bachelor of Technology" in "Mechanical Engineering" during the academic year 2026 – 2027.

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Examiners

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We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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We also thank our faculty members for their suggestions during model reviews.

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ABSTRACT

Solar energy is one of the most promising renewable energy sources because it is clean, abundant and environmentally friendly. Solar thermal collectors are used to convert solar radiation into useful thermal energy for different domestic and industrial applications. Among various solar thermal systems, a Parabolic Trough Collector (PTC) is an important concentrating solar collector that uses a parabolic-shaped reflective surface to concentrate solar radiation onto a receiver tube containing a working fluid.

The present project, entitled “Design and Optimization of Solar Parabolic Trough Collector,” focuses on the experimental investigation of a fabricated solar parabolic trough collector using normal water and saline water as working fluids. The performance of the collector is studied under different flow rates and solar radiation conditions. The experimental investigation includes flow rates of 9 litres per minute (LPM) and 26 litres per minute under full and half solar radiation conditions. Both normal water and saline water are tested to understand the effect of working fluid, flow rate and available solar radiation on the thermal performance of the collector.

The experimental setup consists of a parabolic reflective surface, a receiver tube, a fluid circulation arrangement and the necessary measuring instruments. Readings related to the operating conditions and fluid temperatures are recorded during the experiments. These observations are used to compare the thermal behaviour of the collector under different experimental conditions. The useful heat gain, thermal efficiency and other relevant performance parameters can be evaluated from the recorded data wherever the required measurements are available.

The main objective of this project is to identify the operating conditions that provide better thermal performance and to understand the influence of normal water, saline water, flow rate and solar radiation on the heat transfer process. The experimental results can help in evaluating the suitability of the collector for small-scale solar thermal applications. The conclusions and optimization recommendations are based on the actual experimental observations obtained from the fabricated model.

Keywords: Solar energy, Parabolic Trough Collector, Normal Water, Saline Water, Mass Flow Rate, Solar Radiation, Heat Transfer, Thermal Efficiency.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
PTC	Parabolic Trough Collector
LPM	Litres Per Minute
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
HTF	Heat Transfer Fluid
DNI	Direct Normal Irradiance
Q	Heat Transfer Rate
PFD	Parabolic Focal Distance
SPT	Solar Parabolic Trough
Tin	Inlet Temperature
Tout	Outlet Temperature
ANSYS	Analysis System software
SI	International System of Units

LIST OF SYMBOLS

Symbol	Description	Unit
(Qu)	Useful heat gain	W
(m)	Mass flow rate	kg/s
(Cp)	Specific heat capacity	J/kg.K
(T{in)	Inlet fluid temperature	°C
(Tout)	Outlet fluid temperature	°C
(**ΔT**)	Temperature difference	K
(G)	Solar irradiance	W/m²
(Aa)	Collector aperture area	M ²
(n)	Thermal efficiency	%
(p)	Fluid density	kg/m³
(V)	Fluid volume	m³
(t)	Time	s
(V)	Volumetric flow rate	m³/s
(W)	Watt, unit of power	W
(L)	Length	m
(D)	Diameter	m

CHAPTER 1

INTRODUCTION

1.1 Introduction to Solar Energy

Energy is one of the basic requirements for the development of modern society. It is required for domestic activities, transportation, agriculture, manufacturing and various industrial processes. At present, a large portion of the world's energy demand is fulfilled by conventional sources such as coal, petroleum and natural gas. However, these sources are limited in availability and their continuous use contributes to air pollution and greenhouse gas emissions.

Solar energy is a renewable source of energy obtained from the Sun. It is avail free in nature and it can be used for to generate electricity. Solar energy can also reduce consumtion of fuels and also develop clean energy systems.

Solar energy it’s a technology that can be classified into solar PV systems & solar thermal systems. Solar thermal technology is useful in applications such as water heating, drying, cooking, industrial heating and power generation.

The amount of solar energy available at a particular location depends on factors such as geographical position, season, time of day, cloud cover and atmospheric conditions. Therefore, the performance of a solar thermal system is influenced by the intensity of solar radiation and the design of the collector.

In the present project, solar energy is utilized through a Parabolic Trough Collector (PTC). The collector concentrates solar radiation onto a receivertube through which a working fluid flows. The heat transferred to the fluid is studied under different experimental conditions using normal water and saline water at different flow rates.

1.2 Solar Thermal Energy

Solar thermal energy refers to the heat obtained by absorbing solar radiation. A solar thermal system consists of components that collect sunlight, convert it into heat and transfer that heat to a suitable working medium.

The working principle of a solar thermal collector is based on the absorption of solar radiation by a receiving surface. When solar radiation strikes the absorber, a portion of the incident energy is converted into thermal energy. This heat can then be transferred to water, oil, air or another suitable working fluid.

Solar thermal systems can be divided into non-concentrating and concentrating collectors. Non concentrate collector receive solar radiation in a close area . It is a collectors use optical surfaces to direct solar radiation from a larger to smaller receive area.

A parabolic trough collector uses a curved reflective surface to concentrate sunlight on a receiver tube. The fluid flowing through the tube absorbs the heat and transfers it for further use or storage.

The thermal performance of a solar collector depends on several factors, including solar irradiance, reflective surface quality, receiver characteristics, fluid properties, flow rate and heat losses to the surroundings.

1.3 Solar Collectors

A solar collector is a device that uses sunlight to generate heat. This heat can be used for heating water, industrial processes and electricity generation. Different types of solar collectors are available, and each type is designed to meet specific heating requirements.

1. Flat Plate Collector

A flat plate is use to collects the heat, when the plate is heat by the solar energy it helps to heat the water. There are different types of collectors are available in market

2. Evacuated Tube Collector

A collector consists of glass tubes use to absorb the solar energy. vacuum helps to reduce heat loss. Heat collected in tubes helps to heat the fluid. and maintain suitable temperature conditions

3. Parabolic Trough Collector

Parabolic trough collector have shape of parabola. It helps to concentrate the sunlight on the receiver tube, and helps to gain the heat

4. Parabolic Dish Collector

A collector shape is like a dish, it focus the sunlight on the focal point and helps to rise the temperature and heat gain

5. Central Receiver Solar System

In this system mirrors are use to concentrate the sunlight on receiver tube. this system is use in electricity generation

For project, we select ptc to study how concentrated solar energy use in heat gain

1.4 Parabolic Trough Collector

Parabolic trough collector is collects the sunlight and heat the fluid. The shape of the collector is parabola . it helps to focus the sunlight on the absorber tube

The main components are pump, reflector, receiver tube, supporting frame and glass tube. reflector is concentrates the sunlight, receiver tube absorbs the diffuse energy and heat the flowing fluid support structure helps to maintain the alignment.

The parabola shape helps to cocentrate the solar radiation. As the fluid moves through the tube, its temperature increases. The amount of heat gained depends on the intensity of sunlight, the flow rate of the fluid and the heat lost to the surroundings.

The performance of a parabolic trough collector depends on the design of the reflector, the position of the receiver tube and the properties of the working fluid. Proper alignment is also important to ensure that the maximum possible sunlight reaches the receiver.

In the present project, the collector is used to study the conversion of solar radiation into useful heat and the transfer of this heat to a flowing liquid. The main components of the collector are explained in the following section.

9 26 LPM

1. Parabolic reflector
2. Receiver tube
3. Supporting structure
4. Fluid circulatory system
5. Pump or motor
6. Flow measure
7. Temperature measure
8. Solar radiate

Basically its driven by the design of reflector determines the how effective solar radiation is towards by the receiver.

The present projects its investigate a parabolic trough collector using normal water and saline water as working fluids and its experimental set up for flow rate at 9 LPM & 26 LPM for half and low radiations for operating better conditions.

1.5 Working Principle of Parabolic Trough Collector

The working principle of a parabolic trough collector is based on the reflection and concentration of solar radiation. A parabolic reflector has a geometrical shape that directs approximately parallel incoming rays towards its focal line when the reflector is correctly aligned with the Sun.

The difference between these temperatures is used to estimate the useful heat gained by the fluid, provided that the mass flow rate and specific heat capacity are known or appropriately determined.

The basic working sequence is as follows:

1. Solar radiation falls on the parabolic reflective surface.
2. The reflect radiation direct towards receiver tube.

3. The receiver absorbs the concentrated solar energy.
4. Heat pass to receiver to the working fluid.
5. Measure data for better collector performance.

Heat transfer rate formula

$Q = \dot{m}C_p(To - Ti)$

Where:

Q_u = useful heat gain rate in watts (W)

$\dot{m}$ = mass flow rate of the working fluid in kilograms per second (kg/s)

C_p = specific heat capacity of the working fluid in joules per kilogram kelvin (J/kg·K)

T_{out} = outlet fluid temperature

T_{in} = inlet fluid temperature

This equation indicates that useful heat gain depends on the mass flow rate, the specific heat capacity of the fluid and the temperature rise across the receiver.

The thermal efficiency of the collector can be evaluated by comparing the useful heat gain with the incident solar power collected over the defined aperture area. The exact calculation requires suitable measurements of solar irradiance and collector aperture area, along with the thermal data.

1.6 Need for the Present Project

The increasing demand for energy and the environmental effects associated with conventional fuels have encouraged the development of renewable energy technologies. Solar thermal systems provide an opportunity to utilize solar radiation directly for heating applications.

However, the actual performance of a solar collector changes with environmental conditions and operating parameters. A change in solar radiation can affect the amount of energy available to the receiver. Similarly, the flow rate influences the time available for heat transfer and the temperature rise of the working fluid.

The selection of a working fluid is also important because different fluids have different thermal properties. These properties can influence the heat absorbed by the fluid and the resulting temperature difference. Therefore, experimental comparison of different working fluids can provide useful information about collector operation.

The present project is undertaken to investigate these effects using a fabricated parabolic trough collector. Normal water and saline water are used as working fluids, while two flow rates, 9 LPM and 26 LPM, are considered. The tests are conducted under full and half solar radiation conditions measured during experimentation.

1.7 Problem Statement

The working of a parabolic trough collector is influenced by several factors, such as the amount of sunlight available, the quality of the reflecting surface, the design of the receiver tube and the type and flow rate of the fluid used. These factors determine how effectively solar energy is converted into heat and transferred to the fluid.

The flow rate is an important factor in this process. When the fluid flows faster through the receiver tube, a larger quantity of fluid is heated over a given period, but its temperature may increase by a smaller amount. When the flow is slower, the fluid may gain more temperature as it passes through the tube.

Normal & saline water is use to study. The addition of salt changes some of the physical properties of water, which may affect its heating behaviour. The difference in their performance cannot be determined only by theoretical assumptions. Therefore, both fluids should be tested under similar operating conditions, and their temperature changes and heat transfer characteristics should be compared using experimental results.

The problem addressed in this project is to evaluate the thermal behaviour of a fabricated solar parabolic trough collector using normal water and saline water at 9 LPM and 26 LPM under full and half measured solar radiation conditions.

1.8 Aim and Objectives

1.8.1 Aim

The main aim of these project is design, fabricate, experiment, and investigation a solar trough collector to evaluate a thermal efficiency for better solar radiations.

1.8.2 Objectives

The major objectives of the project are:

1. To study the design, construction and working principle of a parabolic trough collector.
2. To fabricate and operate a working model of the solar parabolic trough collector.
3. To investigate collector performance using normal water and saline water as working fluids.
4. To conduct experiments at flow rates of 9 LPM and 26 LPM.
5. To evaluate the effect of full and half solar radiation conditions on collector performance.

1.9 Scope of the Project

The present project deals with the design, fabrication and testing of a small-scale solar parabolic trough collector.

For our project we use saline and normal water also test in different flow rates one is 9 lpm and one is 26 lpm. The project is tested in different radiation to find the effect of sunlight

5 thermocouple sensors are use to study the experiment . sensors are fix at inlet outlet glass tube absorber and one is for surrounding ambient temperature.

The project identifies the gap also helps in find the different techniques to increase the efficiency

The main focus is how can increase the heat gain so for that we conduct eight types of different experiments

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Previous studies have also investigated different methods of improving heat transfer, such as modifying the receiver tube, reducing thermal losses and using alternative heat transfer fluids. These studies provide a useful foundation for understanding the thermal behaviour of a solar parabolic trough collector.

The present project focuses on the experimental performance of a fabricated PTC using normal water and saline water. Two volumetric flow rates, 9 LPM and 26 LPM, are considered under full and half solar radiation conditions. The following literature review discusses relevant research and explains how the present experimental investigation relates to earlier work.

2.2 Previous Studies on Solar PTC

2.2.1 Study by Salgado Conrado et al. (2017)

Salgado Conrado, Rodriguez-Pulido and Calderón published a review on the thermal performance of parabolic trough solar collectors in 2017. Their work discussed mathematical models, numerical simulations and experimental methods used to evaluate the performance of PTC systems.

The review explained that collector performance is affected by parameters such as receiver temperature, heat losses, environmental conditions and heat transfer characteristics. It also discussed different approaches used by researchers to improve the thermal performance of solar collectors.

This study is relevant to the present project because it establishes the importance of evaluating a PTC through suitable thermal performance parameters rather than considering only the outlet temperature.

2.2.2 Study by Sandeep and Arunachala (2017)

Sandeep and Arunachala reviewed different techniques for improving heat transfer in solar parabolic trough collectors. Their study discussed the use of improved receiver arrangements, inserts, nanofluids and evacuated receiver tubes.

The authors explained that heat losses from the receiver can reduce the amount of solar energy transferred to the working fluid. They also discussed methods for increasing heat transfer between the receiver surface and the fluid flowing inside it.

The study indicates that receiver design and heat transfer characteristics are important considerations when evaluating the performance of a PTC. In the present project, the

emphasis is on comparing the working fluids and operating conditions using an existing fabricated model.

2.2.3 Study by Kumar and Kumar (2018)

Kumar and Kumar experimentally investigated the thermal performance of a solar parabolic trough collector at different flow rates. Their experimental arrangement used water as the working fluid and considered the effect of operating conditions on water temperature rise, useful heat gain and collector efficiency.

The study reported that the mass flow rate is an important factor affecting collector performance. It also considered the influence of receiver glazing and collector operating mode.

This research is closely related to the present project because two different flow rates, 9 LPM and 26 LPM, are used in the experimental investigation. The results from the present model can be compared in terms of temperature rise, useful heat gain and thermal efficiency, provided that the necessary measurements are available.

However, the results obtained from the earlier study cannot be directly applied to the present model because the collector geometry and experimental conditions may be different.

2.2.4 Study by Bellos et al. and Related Researchers (2018)

Research on the modelling and performance analysis of parabolic trough collectors has examined the optical and thermal processes involved in solar energy collection. A comprehensive review published in *Applied Energy* in 2018 discussed optical models, heat transfer models, simulation techniques and possible methods of performance improvement.

The review highlighted the importance of understanding both the concentration of solar radiation and the thermal behaviour of the receiver. Optical losses can reduce the energy reaching the receiver, while thermal losses reduce the useful heat transferred to the working fluid.

This research provides a theoretical basis for studying the influence of solar radiation and collector design on the performance of the fabricated model used in the present project.

2.2.5 Patel et al. (2022)

The review explained that the performance of a PTC depends on the combined effect of its optical design, thermal properties and operating conditions. The choice of working fluid is also important because its thermal properties influence heat transfer within the receiver.

2.3 Effect of Working Fluid on Heat Transfer

The working fluid is an important part of a solar parabolic trough collector because it absorbs the heat produced at the receiver tube and transfers it to another location. The amount of heat carried by the fluid depends on its flow rate, specific heat capacity, temperature rise and the heat transfer taking place inside the tube.

The water is use because it is easily available and salt is use for increase the density and salt is a cheap source than other nanofluids

Density of both water is different so output will change

Both fluids are tested in different flow conditions

To identifies that actually output is different we records the inlet and outlet temperature

2.4 Effect of Mass Flow Rate

$Q_u = \dot{m}\, C_p\, (T_{out} - T_{in})$

The heat absorb by fluid is increase when the flow is slow and decrease when flow is fast, the fluid may become hotter because it spends more time inside the receiver.

9 LPM and 26 LPM flow rate are used to analyze the performance of different waters use in ptc. The readings are compare to se the heat gain

2.5 Effect of Solar Radiation

Gain of energy is depends on how much solar radiation are falls on plates and diffuse in receiver tube

Surrounding ambient temperature and wind are also effects the heat gain. These factors can change during an experiment and influence the temperature of the fluid. For this reason, solar radiation and other operating conditions are considered when comparing the experimental results.

The present project includes two measured radiation conditions: full radiation and half radiation. These conditions are used to study the thermal response of the collector at the two selected flow rates with both working fluids.

For a meaningful comparison, the radiation measurements should be recorded with their units and the measurement method should be described. If solar radiation changes during a test, the corresponding readings and measurement times should be considered during data analysis.

2.6 Research Gap

These literature review shows that how considerable research has been carried out by design for thermal and modelling finish. We had been learned previously studied that design should be favorable and optimal condition based project modelling system .

The water 9 LPM & 26 LPM based on fabricated model condition for study purpose on half and full radiation on its flow rate

CHAPTER 3

DESIGN AND CONSTRUCTION OF SOLAR PARABOLIC TROUGH COLLECTOR

3.1 Introduction

The design are in parabola shape for collector and its important our project.

For this purpose, a parabolic reflective surface is used to direct the incoming solar radiation towards the focal line of the collector.

In the present project, a working model of a solar parabolic trough collector has been designed and fabricated for experimental investigation. An aluminium sheet reflector is used as the reflective surface. The system also consists of a receiver tube, a supporting structure, a fluid circulation arrangement, a motor and a digital rotameter for flow measurement.

The fabricated model is used to investigate the thermal performance of normal water and saline water at two different flow rates, namely 9 LPM and 26 LPM. Experiments are conducted under full and half solar radiation conditions, which are measured during testing.

This chapter describes the main components, design considerations, material selection, construction and assembly of the experimental setup.

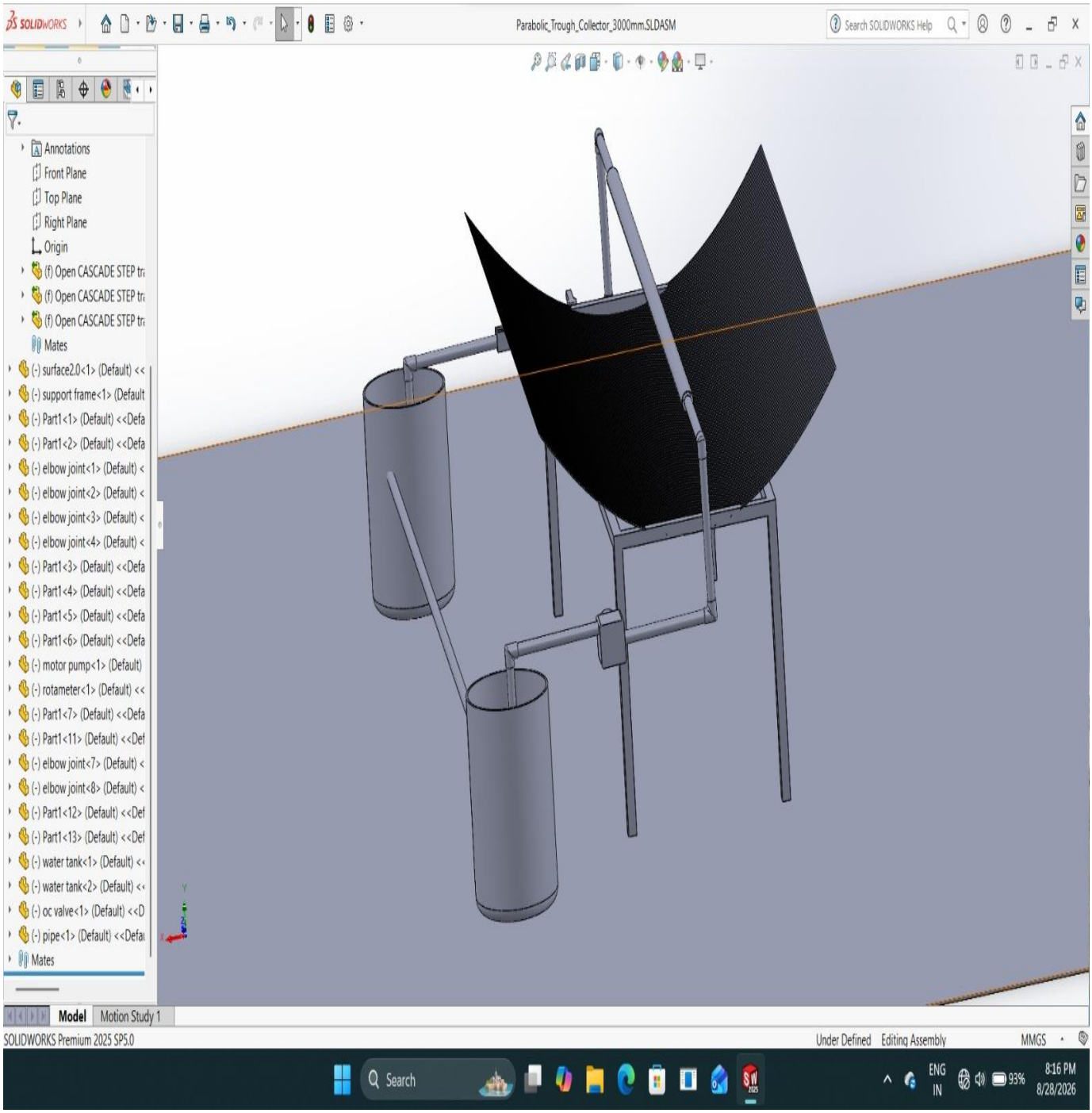
3.2 Basic Design of the Solar Parabolic Trough Collector

A parabolic trough collector consists of a curved reflective surface having a parabolic cross-section. The reflector is designed to direct incident solar radiation towards a receiver tube positioned along its focal line.

The basic design of the present model includes three main sections: the reflector assembly, the receiver assembly and the fluid circulation system. The reflector assembly collects and concentrates solar radiation. The receiver assembly absorbs the concentrated radiation and transfers heat to the working fluid. The circulation system allows the fluid to pass through the receiver at the required flow rate.

The supporting frame holds the reflector and receiver in their required positions. Proper alignment is necessary because a change in the position of the receiver can reduce the amount of reflected radiation reaching its surface.

Figure 3.1: CAD Model of the Solar Parabolic Trough Collector.


SOLIDWORKS
Parabolic_Trough_Collector_3000mm.SLDASM
Search SOLIDWORKS Help
Annotations
Front Plane
Top Plane
Right Plane
Origin
(f) Open CASCADE STEP tr
(f) Open CASCADE STEP tr
(f) Open CASCADE STEP tr
Mates
(-) surface2.0<1> (Default) <<
(-) support frame<1> (Default
(-) Part1<1> (Default) <<Defa
(-) Part1<2> (Default) <<Defa
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(-) water tank<1> (Default) <<
(-) water tank<2> (Default) <<
(-) oc valve<1> (Default) <<D
(-) pipe<1> (Default) <<Defa
Mates
Model
Motion Study 1
SOLIDWORKS Premium 2025 SP5.0
Under Defined
Editing Assembly
MMGS
Search
ENG
IN
93%
8:16 PM
8/28/2026

3.3 Parabolic Reflector

So these parabola reflect for its our own component function based on aperature area and reflect the radiation based on the structure optimization purpose.

Its suitability as a reflector depends on the surface finish, reflectivity, shape accuracy and durability under outdoor conditions.



3.3.1 Geometry of the Parabolic Reflector

So the equation are
Y = x^2 /4f

Where:

- (x) = horizontal distance from the vertex of the parabola, in metres
- (y) = vertical distance from the vertex, in metres
- (f) = focal length of the parabola, in metres

3.4 Receiver Tube

The receiver tube is placed along the focal line of the parabolic reflector, where it receives concentrated sunlight. The tube absorbs the solar energy and transfers the heat to the working fluid flowing inside it. As the fluid passes through the tube, its temperature increases. Proper positioning of the receiver is important to ensure that maximum sunlight reaches its surface. . During the experiment, the tube must be firmly supported and its connections should be checked to prevent leakage and maintain smooth fluid flow.



3.4.1 Functions of the Receiver Tube

The main important part is

1. To provide a passage for continuous fluid circulation.
2. To allow measurement of the inlet and outlet fluid temperatures.
3. To support the experimental evaluation of heat gain.

3.5 Working Fluid Circulation System

The working fluid circulation system transports the fluid through the receiver tube. It is an important part of the experimental setup because the flow rate affects the temperature rise and the quantity of heat carried by the fluid.

In the present project, normal water and saline water are used as the working fluids. The fluid is circulated through the experimental setup at volumetric flow rates of 9 LPM and 26 LPM.

The circulation arrangement includes the fluid supply, connecting pipes or hoses, motor-driven circulation equipment as fitted in the actual model, the receiver tube and the return or collection arrangement. The precise arrangement should be represented using the actual experimental setup.

During operation, the fluid enters the receiver at the inlet temperature, absorbs heat as it flows through the receiver and leaves at the outlet temperature. These temperatures are recorded for thermal analysis.

3.5.1 Normal Water

Normal water is easily available and use as working fluid to identifies the heat gain

3.5.2 Saline Water

For testing that density affects the heat gain or not we use saline water as second fluid .the salt is also one of available source

3.6 Motor and Flow Measurement Arrangement

3.6.1 Motor

A motor is use for circulate the water from the system. The motor is uses in the project is generate 0.5 hp

The max flow rate is about 26 lpm

To reduce the flow rate from 26 lpm to 9 lpm we use the 2 open close valve, one at inlet and one at just before the rotameter

This helps to change the flow rate of the water

3.6.2 Digital Rotameter

A digital rotameter or digital flow-measuring instrument is used to measure the volumetric flow rate of the working fluid.

When the water flows from rotameter it calculates the volumetric flow rate of water

It gives reading of total flow from cycle start to end and also how much of water transfer in one minute. It measures in lpm

mass flow rate = density * volume flow

Where:

- (m) = mass flow rate, in kg/s
- (p) = density of the working fluid, in kg/m^3
- (V) = volumetric flow rate, in m^3/s

Lpm = 10^ -3 / 60 m^3/s

Water Sparlet
DIGITAL FLOW METER
ADHESIVE

3.7 Material Selection

We use thin aluminium sheets because of its high reflectivity. Also aluminium is corrosion resistant. aluminium sheet can change the shape easily

We use saline water so corrosivity management is very important

We use sealing at both side to create a vaccum.

Table 3.1: Main Components and Material Selection

Sr. No.	Component	Material or specification	Function
1	Reflector	Al 0.5mm	Receives solar heat and transfers it to the fluid.
2	Reciever Tubes	Glass	Holds the collector and keeps its parts in place.
3	Supporting frame	Rod	Carry the working fluid through the system.
4	Connecting pipes or hoses	PVC	Runs the connected fluid circulation equipment
5	Motor	0.5 HP	Measures flow rate of the fluid.
6	Flow instrument	Digital rotameter	Flow measure
7	Temperature instruments	Thermocouple sensor	Temperature data
8	Radiation measuring instrument	Pyranometer	Measures the intensity of sunlight during testing.
9	Fasterner & Joints	Plastic	Join the different parts and keep them securely connected.

3.8 Design Parameters and Dimensions

The size of the collector is determined by its dimensions which affect its optical and thermal performance. The width and the length of the reflector define the reflector aperture area and the position of the receiver is defined by the parabolic shape and the focal length.

Available heat transfer surface depends on the diameter and length of the receiver. The spacing of reflector and receiver is also significant as the receiver should be placed near the focal line.

These design parameters should be taken from the CAD drawing or directly measured from the fabricated model.

Table 3.2: Design Specifications of the Solar PTC

Sr. No.	Design parameter	Symbol	Actual value
1	Reflector material	—	Aluminium sheet
2	Reflector sheet thickness	(t)	0.5 mm
3	Reflector aperture width	(W)	Enter measured value
4	Reflector length	(L)	Enter measured value
5	Focal length	(f)	Enter CAD value
6	Receiver tube material	(Rt)	Enter actual material
7	Receiver tube outer diameter	(Do)	Enter measured value
8	Receiver tube length	(Lr)	Enter measured value
9	Motor rated power	(P)	0.5 HP
10	Lower flow rate	(V1)	9 LPM
11	Higher flow rate	(V2)	26 LPM
12	Working fluids	(Wf)	Normal water and saline water

3.8.1 Aperture Area

Aa = Wa * L c

Where:

- (Aa) = aperture area, in m²
- (Wa) = projected aperture width, in m
- (Lc) = collector length, in m

- The size of the actual model in the CAD should be used for the calculation. The total sheet area is NOT necessarily equal to the aperture area, especially if there are multiple sheets overlapping or a curved reflector.
- The aperture area is needed in the calculation of the incident solar power to the collector.

3.9 Fabrication and Assembly

Fabrication:Turning designed parts into an experimental model that is actually made in a real laboratory.

During the construction process, it is required to follow the approved drawing and keep the required position of the reflector, receiver and circulation arrangement.

The following is a description of the main fabrication and assembly tasks.

Step 1: Preparation of the design

The basic dimensions and arrangement of the reflector, receiver tube and supporting structure are determined from the design drawing. Prior to fabrication, the focal position and clearances are checked.

Step 2: Preparation of the reflector

The aluminium sheet is cut and shaped to the desired dimensions. The reflector is bent in the desired parabolic shape, as closely as possible. The surface is examined for damage, unwanted deformities and irregularities.

The third step in the process is to fabricate the supporting structure.The third step is to fabricate supporting structure.

The supporting frame is built to support the reflector in its desired location. It should offer adequate stability when being installed and during the experiments.

Step 4: Installation of the receiver tube

The receiver tube is positioned on the desired focal line of the reflector. It is aligned to ensure that solar radiation is as effective as the design allows it to be when it reflects off of it.

The fluid circulation arrangement is installed

step 5.

The fluid circulation system is connected to the receiver tube via the pipes, hoses and fittings included in the actual setup. Checking for leakages and direction of flow.

The measuring instruments are installed in

Step 6.

Digital rotameter and measuring devices of temperature and solar radiation are installed or positioned as per the experimental setup. They read and cover the units prior to a test.
Step 7: Inspection and testing.

The put together system is evaluated for structural stability, alignment of receivers, fluid leaks and secure electrical connections. The circulation system is controlled and flow rate is manipulated to the desired test condition.

Step 8: Experimental readiness

Measuring instruments are tested before solar testing and pre-testing operating conditions are noted. The system is then employed for experiments with normal water and saline water at 9 LPM and 26 LPM under the following radiation conditions.

The fabrication process outlined above may be tailored to the actual order of fabrication used during the building of the project.

3.9.1 Final Inspection

Prior to taking experimental readings the following checks should be made:
Reflector must be firmly in place.

2. Receiver tube must be properly set-up.
4. There should be no leakage in the fluid connections.
5. The digital rotameter should give consistent flow values.
6. The temperature and radiation sensors must be in place for taking measurements.
7. The motor and electrical connections should be safe for operation.
8. Collector should be oriented in proper direction with the sun's radiation.
9. The experimental conditions should be recorded consistently.

CHAPTER 4

EXPERIMENTAL SETUP AND METHODOLOGY

4.1 Introduction

It is important that the experimental study be carried out as part of the project since it will allow us to understand the actual performance of the Solar Parabolic Trough Collector (PTC). The objective of this experiment is to understand the influence of the fluid flow rates, working fluid type and the solar radiation on the thermal efficiency of the collector.

Here, normal water and saline water are used as working fluid in this project. Two flow rates (9 and 26 LPM) and two solar radiation levels (full and half) are used in the experiments. The values of the readings taken in the experiments are then compared to determine the performance of the collector under various operating conditions.

4.2 Experimental Setup

The experimental setup includes a parabolic reflector, a receiver tube, a supporting frame for the setup, a fluid circulation system, a motor, a digital rotameter and temperature measuring instruments.

The reflector concentrates the incoming solar radiation onto the receiver tube placed near its focal line.

The working fluid flows through the receiver tube and absorbs heat from the concentrated solar energy. The fluid circulation system maintains the required flow rate during the experiment. The digital rotameter is used to measure the flow rate, while the temperature and solar radiation readings are recorded for further analysis.

The experimental set-up is designed to compare the thermal performance of the collector when employing different fluids, different flow rates and different conditions of solar radiation.

4.3 Instruments and Components Used

The main components and instruments used in the experimental setup are listed below.

Sr. No.	Component / Instrument	Purpose
1	Al reflect sheet	Reflector sheet and absorbs
2	Receiver tube	Absorb heat
3	Supporting structural frame	Supporting apparatus
4	Motor (0.5 HP)	Standard rating
5	Digital rotameter	Flow measure
6	Temperature-measuring instrument	reading
7	Solar radiation measuring instrument	reading
8	Pipes and connecting tubes	Joints

9	Water container or reservoir	flow

Note: Keep only those instruments and components that are actually present in your model. Add the exact instrument names and specifications wherever available.

4.4 Working Fluids

In the present experimental investigation two working fluid is used, normal water and saline water.

1. Normal Water: Normal water is used as the reference working fluid. The heat absorbed by the fluid is measured by the temperature rise of the fluid during circulation in the receiver tube.

2. Saline Water: Saline water is water which has been made salty using salt. Tested under similar experimental conditions to compare it with normal water. In all the saline-water tests the concentration of the saline should be maintained at the same value and should be stated in the report.

The specific heat capacity and density of the fluids can have an impact on the heat gain calculation. So, the same calculation method should be applied for both fluids and the appropriate properties should be used for the same temperature and concentration.

4.5 Experimental Variables

The operating parameters which vary in the experiment are the working fluid, flow rate and level of solar radiation.

- Process: Mixing of normal and saline water.
- Flow rate: 9 LPM and 26 LPM.

Solar radiation: Full radiation and half radiation, as determined by the measurements taken during testing.

The other factors (collector arrangement, receiver tube and measurement procedure) should be as uniform as possible to be able to compare the conditions.

4.6 Full and Half Solar Radiation Conditions

The collector is tested under two solar radiation conditions.

Full Solar Radiation: The collector is exposed to the available solar radiation under the selected full-radiation test condition. The actual radiation value is measured and recorded during the experiment.

Half Solar Radiation: The collector is tested under a lower radiation condition corresponding to the measured half-radiation test. The actual radiation value is recorded rather than assuming that it is exactly half of the full-radiation value.

Solar radiation can change with time, cloud cover and the position of the sun. Therefore, radiation readings should be recorded alongside the temperature readings so that the results can be compared correctly.

4.7 Experimental Procedure

The following procedure is used to carry out the experiment:

1. The complete Solar Parabolic Trough Collector setup is inspected before starting the experiment.
2. The reflector surface and receiver tube are checked to ensure that they are properly positioned.
3. The selected working fluid, either normal water or saline water, is filled into the circulation system.
4. The pump or motor is operated according to the arrangement of the experimental setup.
5. The digital rotameter is used to set and check the required flow rate.
6. The experiment is conducted at a flow rate of 9 LPM, and the relevant temperature and solar radiation readings are recorded.
7. The same procedure is repeated at a flow rate of 26 LPM.
8. The experiment is repeated under full and half solar radiation conditions.
9. The working fluid is changed from normal water to saline water, and the same set of tests is performed.
10. All observations are entered into the experimental observation table for calculation and comparison.

Where possible, readings should be taken after the operating conditions become sufficiently stable. The time of measurement should also be recorded.

4.8 Recording of Experimental Readings

The readings obtained during the experiment are used to calculate the heat gain and thermal efficiency of the collector. The following information should be recorded for each test:

- Type of working fluid
- Flow rate in LPM
- Measured solar radiation (W/m2)
- T_1 (°C) - inlet fluid temperature.
- Temperature of outlet fluid °C
- Temperature of the surroundings in °C, if measured
- Time & date of conducting experiment
- You can use the following observation table in the report:

Table 4.1: Experimental test conditions

Record the actual readings for each test in a separate observation table, as shown below.

Parameter	Symbol	Unit
Inlet temperature	Tin	°C
Temp out	To	°C
S R	R	W/m²
F F R	$\dot{v}$	LPM
Ambient temperature	Ta	°C

Note: Enter only measured values in the final observation tables. Do not use assumed readings in place of experimental data.

4.9 Precautions During Experimentation

1. The reflector and receiver tube should be properly aligned.
2. The pipes and connections should be checked for leakage before starting the experiment.
3. The digital rotameter should be checked to ensure that it displays the correct flow rate.
4. The temperature-measuring instruments should be placed correctly.
5. The measured solar radiation should be recorded with its corresponding temperature readings.
6. The working fluid and flow rate should be changed carefully between tests.
7. The same measurement procedure should be followed for normal water and saline water.
8. The collector should be handled carefully because the receiver tube and working fluid may become hot.
9. The actual operating conditions and any unusual observations should be noted in the experiment record.

- Now readings of 9 LPM & 26 LPM of both half and full radiations between Normal & Saline Water. Here are readings we are taken from datalogger via using 5 thermocouple sensors:-

1) Inlet Temperature.
2) Outlet Temperature.
3) T glass.
4) T absorber.
5) T ambient.

Table 1} 9 LPM Full Radiation at Normal Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
21-9-2026	10:00:00	20	22	32	31	21	398	9LPM
21-9-2026	10:15 AM	20	22	32	31	21	404	9LPM
21-9-2026	10:30 AM	20	22	32	31	21	409	9LPM
21-9-2026	10:45 AM	21	23	33	32	22	414	9LPM
21-9-2026	11:00 AM	21	23	33	32	22	430	9LPM
21-9-2026	11:15 AM	21	23	33	32	22	450	9LPM
21-9-2026	11:30 AM	22	24	34	33	23	478	9LPM
21-9-2026	11:45 AM	22	24	34	33	23	490	9LPM
21-9-2026	12:00 PM	22	24	34	33	24	489	9LPM
21-9-2026	12:15 PM	23	25	35	34	24	509	9LPM
21-9-2026	12:30 PM	23	25	35	34	24	523	9LPM
21-9-2026	12:45 PM	23	25	35	34	26	538	9LPM
21-9-2026	1:00 PM	23	25	35	34	24	546	9LPM
21-9-2026	1:15 PM	24	26	36	35	25	546	9LPM
21-9-2026	1:30 PM	24	26	36	35	25	550	9LPM
21-9-2026	1:45 PM	25	27	37	36	27	556	9LPM
21-9-2026	2:00 PM	25	27	37	36	26	559	9LPM

21-9-2026	2:15 PM	27	29	39	38	28	567	9LPM
21-9-2026	2:30 PM	27	29	39	38	28	567	9LPM
21-9-2026	2:45 PM	28	30	40	39	30	570	9LPM
21-9-2026	3:00 PM	29	31	41	40	30	586	9LPM
21-9-2026	3:15 PM	29	31	41	40	30	591	9LPM
21-9-2026	3:30 PM	29	31	41	40	31	602	9LPM
21-9-2026	3:45 PM	28	30	40	39	29	604	9LPM
21-9-2026	4:00 PM	28	30	40	39	29	601	9LPM
21-9-2026	4:15 PM	26	28	38	37	27	593	9LPM
21-9-2026	4:30 PM	25	27	37	36	26	513	9LPM
21-9-2026	4:45 PM	23	25	35	34	24	434	9LPM
21-9-2026	5:00 PM	23	25	35	34	24	397	9LPM

Table 2} 26 LPM Full Radiation at Normal Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
22-09-26	10:00:00	19	19	30	28	20	400	26LPM
22-09-26	10:15 AM	19	19	30	28	20	402	26LPM
22-09-26	10:30 AM	20	20	31	29	21	404	26LPM
22-09-26	10:45 AM	20	20	31	29	21	405	26LPM
22-09-26	11:00 AM	20	20	31	29	21	409	26LPM
22-09-26	11:15 AM	20	20	31	29	21	415	26LPM
22-09-26	11:30 AM	21	21	32	30	22	420	26LPM
22-09-26	11:45 AM	21	21	32	30	22	423	26LPM
22-09-26	12:00 PM	21	22	32	31	22	429	26LPM
22-09-26	12:15 PM	21	22	32	31	22	436	26LPM

22-09-26	12:30 PM	21	22	33	31	22	450	26LPM
22-09-26	12:45 PM	22	23	33	32	23	470	26LPM
22-09-26	1:00 PM	22	23	33	32	23	483	26LPM
22-09-26	1:15 PM	22	23	33	32	23	485	26LPM
22-09-26	1:30 PM	23	24	34	33	24	493	26LPM
22-09-26	1:45 PM	23	24	34	33	24	510	26LPM
22-09-26	2:00 PM	23	24	34	33	24	528	26LPM
22-09-26	2:15 PM	23	24	34	33	24	550	26LPM
22-09-26	2:30 PM	24	25	35	34	25	545	26LPM
22-09-26	2:45 PM	24	25	35	34	25	533	26LPM
22-09-26	3:00 PM	24	25	35	34	25	544	26LPM
22-09-26	3:15 PM	23	24	34	33	24	553	26LPM
22-09-26	3:30 PM	23	24	34	33	24	529	26LPM
22-09-26	3:45 PM	23	24	34	33	24	442	26LPM
22-09-26	4:00 PM	22	23	33	32	23	422	26LPM
22-09-26	4:15 PM	22	22	32	31	23	398	26LPM
22-09-26	4:30 PM	22	21	31	30	23	368	26LPM
22-09-26	4:45 PM	21	21	31	30	22	302	26LPM
22-09-26	5:00 PM	21	21	31	30	22	250	26LPM

Table 3} 9 LPM Low Radiation at Normal Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
24-09-26	10:00:00	19	19	24	23	20	75	9LPM
24-09-26	10:15 AM	19	19	24	23	20	78	9LPM
24-09-26	10:30 AM	19	19	24	23	20	82	9LPM
24-09-26	10:45 AM	20	20	25	24	21	79	9LPM
24-09-26	11:00 AM	20	20	25	24	21	81	9LPM
24-09-26	11:15 AM	21	21	26	25	22	80	9LPM

24-09-26	11:30 AM	21	21	26	25	22	86	9LPM
24-09-26	11:45 AM	21	22	26	25	22	88	9LPM
24-09-26	12:00 PM	22	22	27	26	23	85	9LPM
24-09-26	12:15 PM	22	22	27	26	23	87	9LPM
24-09-26	12:30 PM	22	22	27	26	23	89	9LPM
24-09-26	12:45 PM	22	22	27	26	23	90	9LPM
24-09-26	1:00 PM	22	22	27	26	23	86	9LPM
24-09-26	1:15 PM	23	24	28	27	25	84	9LPM
24-09-26	1:30 PM	23	24	28	27	25	86	9LPM
24-09-26	1:45 PM	23	24	28	27	24	88	9LPM
24-09-26	2:00 PM	23	24	28	27	24	82	9LPM
24-09-26	2:15 PM	23	23	28	27	25	83	9LPM
24-09-26	2:30 PM	23	23	28	27	26	82	9LPM
24-09-26	2:45 PM	22	22	27	26	23	84	9LPM
24-09-26	3:00 PM	22	22	27	26	23	85	9LPM
24-09-26	3:15 PM	22	21	27	26	24	84	9LPM
24-09-26	3:30 PM	21	21	26	25	24	85	9LPM
24-09-26	3:45 PM	21	21	26	25	22	84	9LPM
24-09-26	4:00 PM	21	21	26	25	22	79	9LPM
24-09-26	4:15 PM	21	21	26	25	22	75	9LPM
24-09-26	4:30 PM	21	21	26	25	22	72	9LPM
24-09-26	4:45 PM	21	21	26	25	22	74	9LPM
24-09-26	5:00 PM	21	21	26	25	22	73	9LPM

Table 4} 26 LPM Low Radiation at Normal Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
28-09-26	10:00:00	19	19	23	22	20	75	26 LPM
28-09-26	10:15:00	19	19	23	22	20	75	26 LPM
28-09-26	10:30:00	19	19	23	22	20	76	26 LPM
28-09-26	10:45:00	20	20	24	23	21	76	26 LPM
28-09-26	11:00:00	20	20	24	23	21	76	26 LPM
28-09-26	11:15:00	20	20	24	23	21	75	26 LPM
28-09-26	11:30:00	20	20	24	23	21	78	26 LPM
28-09-26	11:45:00	20	20	24	23	21	78	26 LPM
28-09-26	12:00:00	20	20	24	23	21	79	26 LPM
28-09-26	12:15:00	21	21	25	24	22	81	26 LPM
28-09-26	12:30:00	21	21	25	24	22	83	26 LPM
28-09-26	12:45:00	22	22	26	25	23	82	26 LPM
28-09-26	13:00:00	22	22	26	25	23	83	26 LPM
28-09-26	13:15:00	22	22	26	25	23	85	26 LPM
28-09-26	13:30:00	22	22	26	25	23	85	26 LPM
28-09-26	13:45:00	23	23	27	26	24	85	26 LPM
28-09-26	14:00:00	23	23	27	26	24	83	26 LPM
28-09-26	14:15:00	23	23	27	26	24	82	26 LPM

28-09-26	14:30:00	23	23	27	26	25	82	26 LPM
28-09-26	14:45:00	23	23	27	26	24	82	26 LPM
28-09-26	15:00:00	22	22	26	25	25	81	26 LPM
28-09-26	15:15:00	22	22	26	25	24	80	26 LPM
28-09-26	15:30:00	22	22	26	25	24	79	26 LPM
28-09-26	15:45:00	21	21	25	24	22	79	26 LPM
28-09-26	16:00:00	21	21	25	24	22	76	26 LPM
28-09-26	16:15:00	20	20	24	23	21	75	26 LPM
28-09-26	16:30:00	20	20	24	23	21	74	26 LPM
28-09-26	16:45:00	20	20	24	23	21	74	26 LPM
28-09-26	17:00:00	20	20	24	23	21	74	26 LPM

Table 5} 9 LPM Full Radiation at Saline Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
30-09-26	10:00:00	20	21	31	30	21	411	9LPM
30-09-26	10:15:00	20	21	31	30	21	440	9LPM
30-09-26	10:30:00	20	21	31	30	21	445	9LPM
30-09-26	10:45:00	20	21	31	30	21	472	9LPM
30-09-26	11:00:00	21	22	32	31	22	500	9LPM
30-09-26	11:15:00	21	22	32	31	22	540	9LPM
30-09-26	11:30:00	22	22	32	31	23	545	9LPM
30-09-26	11:45:00	22	23	33	32	23	548	9LPM
30-09-26	12:00:00	23	24	34	33	24	550	9LPM
30-09-26	12:15:00	23	24	34	33	24	552	9LPM
30-09-26	12:30:00	23	24	34	33	24	560	9LPM
30-09-26	12:45:00	24	25	35	34	25	575	9LPM
30-09-26	13:00:00	24	25	35	34	25	583	9LPM
30-09-26	13:15:00	25	26	36	35	26	588	9LPM
30-09-26	13:30:00	25	27	37	36	27	596	9LPM
30-09-26	13:45:00	25	27	37	36	26	602	9LPM
30-09-26	14:00:00	25	27	37	36	28	612	9LPM
30-09-26	14:15:00	25	27	37	36	26	625	9LPM
30-09-26	14:30:00	25	27	37	36	27	628	9LPM
30-09-26	14:45:00	24	25	35	34	25	608	9LPM
30-09-26	15:00:00	24	25	35	34	26	598	9LPM
30-09-26	15:15:00	24	25	35	34	25	556	9LPM
30-09-26	15:30:00	24	25	35	34	25	525	9LPM
30-09-26	15:45:00	23	24	34	33	24	480	9LPM
30-09-26	16:00:00	23	24	34	33	24	410	9LPM
30-09-26	16:15:00	23	24	34	33	24	380	9LPM
30-09-26	16:30:00	23	24	34	33	24	298	9LPM
30-09-26	16:45:00	22	23	33	32	23	266	9LPM
30-09-26	17:00:00	22	23	33	32	23	263	9LPM

Table 6} 26 LPM Full Radiation at Saline Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
3/10/2026	10:00:00	21	21	30	30	22	410	26LPM
3/10/2026	10:15:00	21	21	31	30	22	420	26LPM
3/10/2026	10:30:00	21	21	31	30	22	428	26LPM
3/10/2026	10:45:00	21	22	32	31	22	440	26LPM
3/10/2026	11:00:00	22	23	33	32	23	445	26LPM
3/10/2026	11:15:00	22	23	33	32	23	462	26LPM
3/10/2026	11:30:00	22	23	33	32	23	466	26LPM
3/10/2026	11:45:00	23	24	34	33	24	480	26LPM
3/10/2026	12:00:00	23	24	34	33	24	494	26LPM
3/10/2026	12:15:00	24	25	35	34	25	505	26LPM
3/10/2026	12:30:00	24	25	35	34	26	516	26LPM
3/10/2026	12:45:00	24	25	35	34	27	524	26LPM
3/10/2026	13:00:00	24	25	35	34	25	543	26LPM
3/10/2026	13:15:00	25	26	36	35	26	576	26LPM
3/10/2026	13:30:00	25	26	36	35	27	597	26LPM
3/10/2026	13:45:00	25	26	36	35	28	603	26LPM
3/10/2026	14:00:00	26	27	37	36	27	624	26LPM
3/10/2026	14:15:00	26	27	37	36	27	636	26LPM
3/10/2026	14:30:00	25	26	36	35	26	643	26LPM
3/10/2026	14:45:00	25	26	36	35	26	631	26LPM
3/10/2026	15:00:00	24	25	35	34	25	625	26LPM
3/10/2026	15:15:00	24	25	35	34	25	580	26LPM
3/10/2026	15:30:00	24	25	35	34	26	549	26LPM
3/10/2026	15:45:00	24	25	35	34	25	490	26LPM
3/10/2026	16:00:00	23	24	34	33	24	451	26LPM
3/10/2026	16:15:00	23	24	34	33	24	401	26LPM
3/10/2026	16:30:00	23	23	33	32	24	374	26LPM
3/10/2026	16:45:00	23	23	33	32	24	349	26LPM
3/10/2026	17:00:00	22	22	32	31	23	323	26LPM

Table 7} 9 LPM Low Radiation at Saline Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
5/10/2026	10:00:00	19	19	23	22	20	73	9LPM
5/10/2026	10:15:00	19	19	23	22	20	74	9LPM
5/10/2026	10:30:00	19	19	23	22	20	74	9LPM
5/10/2026	10:45:00	20	20	24	23	21	74	9LPM
5/10/2026	11:00:00	20	20	24	23	21	75	9LPM
5/10/2026	11:15:00	20	20	24	23	21	77	9LPM
5/10/2026	11:30:00	20	20	24	23	21	78	9LPM

5/10/2026	11:45:00	20	20	24	23	21	78	9LPM
5/10/2026	12:00:00	21	21	25	24	22	79	9LPM
5/10/2026	12:15:00	21	21	25	24	23	79	9LPM
5/10/2026	12:30:00	21	21	25	24	23	79	9LPM
5/10/2026	12:45:00	21	22	25	24	22	80	9LPM
5/10/2026	13:00:00	22	22	26	25	23	80	9LPM
5/10/2026	13:15:00	22	22	26	25	24	81	9LPM
5/10/2026	13:30:00	23	23	27	26	24	82	9LPM
5/10/2026	13:45:00	23	23	27	26	24	82	9LPM
5/10/2026	14:00:00	24	24	28	27	25	82	9LPM
5/10/2026	14:15:00	24	25	28	27	26	83	9LPM
5/10/2026	14:30:00	25	25	29	28	26	83	9LPM
5/10/2026	14:45:00	25	25	29	28	26	84	9LPM
5/10/2026	15:00:00	24	25	28	27	25	85	9LPM
5/10/2026	15:15:00	24	25	28	27	27	87	9LPM
5/10/2026	15:30:00	23	24	27	26	27	84	9LPM
5/10/2026	15:45:00	23	23	27	26	25	83	9LPM
5/10/2026	16:00:00	23	23	27	26	24	82	9LPM
5/10/2026	16:15:00	22	22	26	25	23	81	9LPM
5/10/2026	16:30:00	22	22	26	25	23	79	9LPM
5/10/2026	16:45:00	21	21	25	24	22	78	9LPM
5/10/2026	17:00:00	21	21	25	24	22	77	9LPM

Table 8} 26 LPM Low Radiation at Saline Water.

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
6/10/2026	10:00:00	19	19	23	22	20	72	26LPM
6/10/2026	10:15:00	19	19	23	22	20	72	26LPM
6/10/2026	10:30:00	19	19	23	22	20	72	26LPM
6/10/2026	10:45:00	20	20	24	23	21	73	26LPM
6/10/2026	11:00:00	20	20	24	23	21	74	26LPM
6/10/2026	11:15:00	20	20	24	23	21	74	26LPM
6/10/2026	11:30:00	21	21	25	24	22	75	26LPM
6/10/2026	11:45:00	21	21	25	24	22	75	26LPM
6/10/2026	12:00:00	21	21	25	24	22	75	26LPM
6/10/2026	12:15:00	21	21	25	24	22	76	26LPM
6/10/2026	12:30:00	22	22	26	25	23	76	26LPM
6/10/2026	12:45:00	22	22	26	25	23	77	26LPM
6/10/2026	13:00:00	22	22	26	25	23	78	26LPM
6/10/2026	13:15:00	22	22	26	25	23	79	26LPM
6/10/2026	13:30:00	23	23	27	26	24	81	26LPM
6/10/2026	13:45:00	23	23	27	26	24	82	26LPM
6/10/2026	14:00:00	23	24	27	26	24	82	26LPM
6/10/2026	14:15:00	23	24	27	26	24	83	26LPM
6/10/2026	14:30:00	24	24	28	27	26	83	26LPM
6/10/2026	14:45:00	24	24	28	27	26	84	26LPM

6/10/2026	15:00:00	24	24	28	27	25	85	26LPM
6/10/2026	15:15:00	23	23	27	26	24	84	26LPM
6/10/2026	15:30:00	23	23	27	26	26	83	26LPM
6/10/2026	15:45:00	23	23	27	26	24	82	26LPM
6/10/2026	16:00:00	22	22	26	25	23	82	26LPM
6/10/2026	16:15:00	21	21	25	24	22	80	26LPM
6/10/2026	16:30:00	21	21	25	24	22	79	26LPM
6/10/2026	16:45:00	20	20	24	23	21	78	26LPM
6/10/2026	17:00:00	20	20	24	23	21	77	26LPM

Table 1} CFD RESULT

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
21-9-2026	10:00:00	20	24	32	31	21	398	9LPM
21-9-2026	10:15 AM	20	24	32	31	21	404	9LPM
21-9-2026	10:30 AM	20	24	32	31	21	409	9LPM
21-9-2026	10:45 AM	21	25	33	32	22	414	9LPM
21-9-2026	11:00 AM	21	25	33	32	22	430	9LPM
21-9-2026	11:15 AM	21	25	33	32	22	450	9LPM
21-9-2026	11:30 AM	22	26	34	33	23	478	9LPM
21-9-2026	11:45 AM	22	26	34	33	23	490	9LPM
21-9-2026	12:00 PM	22	26	34	33	24	489	9LPM
21-9-2026	12:15 PM	23	27	35	34	24	509	9LPM
21-9-2026	12:30 PM	23	27	35	34	24	523	9LPM
21-9-2026	12:45 PM	23	27	35	34	26	538	9LPM
21-9-2026	1:00 PM	23	27	35	34	24	546	9LPM
21-9-2026	1:15 PM	24	27	36	35	25	546	9LPM
21-9-2026	1:30 PM	24	28	36	35	25	550	9LPM
21-9-2026	1:45 PM	25	29	37	36	27	556	9LPM
21-9-2026	2:00 PM	25	29	37	36	26	559	9LPM
21-9-2026	2:15 PM	27	31	39	38	28	567	9LPM
21-9-2026	2:30 PM	27	31	39	38	28	567	9LPM

21-9-2026	2:45 PM	28	32	40	39	30	570	9LPM
21-9-2026	3:00 PM	29	32	41	40	30	586	9LPM
21-9-2026	3:15 PM	29	33	41	40	30	591	9LPM
21-9-2026	3:30 PM	29	33	41	40	31	602	9LPM
21-9-2026	3:45 PM	28	32	40	39	29	604	9LPM
21-9-2026	4:00 PM	28	32	40	39	29	601	9LPM
21-9-2026	4:15 PM	26	30	38	37	27	593	9LPM
21-9-2026	4:30 PM	25	28	37	36	26	513	9LPM
21-9-2026	4:45 PM	23	27	35	34	24	434	9LPM
21-9-2026	5:00 PM	23	26	35	34	24	397	9LPM

Table 2} CFD RESULT

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
22-09-26	10:00:00	19	20	30	28	20	400	26LPM
22-09-26	10:15 AM	19	20	30	28	20	402	26LPM
22-09-26	10:30 AM	20	21	31	29	21	404	26LPM
22-09-26	10:45 AM	20	21	31	29	21	405	26LPM
22-09-26	11:00 AM	20	21	31	29	21	409	26LPM
22-09-26	11:15 AM	20	21	31	29	21	415	26LPM
22-09-26	11:30 AM	21	23	32	30	22	420	26LPM
22-09-26	11:45 AM	21	23	32	30	22	423	26LPM
22-09-26	12:00 PM	21	24	32	31	22	429	26LPM
22-09-26	12:15 PM	21	23	32	31	22	436	26LPM
22-09-26	12:30 PM	21	24	33	31	22	450	26LPM
22-09-26	12:45 PM	22	24	33	32	23	470	26LPM
22-09-26	1:00 PM	22	25	33	32	23	483	26LPM
22-09-26	1:15 PM	22	25	33	32	23	485	26LPM

22-09-26	1:30 PM	23	26	34	33	24	493	26LPM
22-09-26	1:45 PM	23	26	34	33	24	510	26LPM
22-09-26	2:00 PM	23	26	34	33	24	528	26LPM
22-09-26	2:15 PM	23	26	34	33	24	550	26LPM
22-09-26	2:30 PM	24	26	35	34	25	545	26LPM
22-09-26	2:45 PM	24	26	35	34	25	533	26LPM
22-09-26	3:00 PM	24	26	35	34	25	544	26LPM
22-09-26	3:15 PM	23	24	34	33	24	553	26LPM
22-09-26	3:30 PM	23	24	34	33	24	529	26LPM
22-09-26	3:45 PM	23	24	34	33	24	442	26LPM
22-09-26	4:00 PM	22	23	33	32	23	422	26LPM
22-09-26	4:15 PM	22	23	32	31	23	398	26LPM
22-09-26	4:30 PM	22	22	31	30	23	368	26LPM
22-09-26	4:45 PM	21	22	31	30	22	302	26LPM
22-09-26	5:00 PM	21	22	31	30	22	250	26LPM

Table 3} CFD RESULT

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
24-09-26	10:00:00	19	20	24	23	20	75	9LPM
24-09-26	10:15 AM	19	20	24	23	20	78	9LPM
24-09-26	10:30 AM	19	20	24	23	20	82	9LPM
24-09-26	10:45 AM	20	21	25	24	21	79	9LPM
24-09-26	11:00 AM	20	21	25	24	21	81	9LPM
24-09-26	11:15 AM	21	22	26	25	22	80	9LPM
24-09-26	11:30 AM	21	22	26	25	22	86	9LPM
24-09-26	11:45 AM	21	22	26	25	22	88	9LPM
24-09-26	12:00 PM	22	23	27	26	23	85	9LPM
24-09-26	12:15 PM	22	23	27	26	23	87	9LPM
24-09-26	12:30 PM	22	23	27	26	23	89	9LPM
24-09-26	12:45 PM	22	23	27	26	23	90	9LPM
24-09-26	1:00 PM	22	23	27	26	23	86	9LPM
24-09-26	1:15 PM	23	24	28	27	25	84	9LPM
24-09-26	1:30 PM	23	24	28	27	25	86	9LPM
24-09-26	1:45 PM	23	24	28	27	24	88	9LPM
24-09-26	2:00 PM	23	24	28	27	24	82	9LPM

24-09-26	2:15 PM	23	24	28	27	25	83	9LPM
24-09-26	2:30 PM	23	24	28	27	26	82	9LPM
24-09-26	2:45 PM	22	23	27	26	23	84	9LPM
24-09-26	3:00 PM	22	23	27	26	23	85	9LPM
24-09-26	3:15 PM	22	23	27	26	24	84	9LPM
24-09-26	3:30 PM	21	22	26	25	24	85	9LPM
24-09-26	3:45 PM	21	22	26	25	22	84	9LPM
24-09-26	4:00 PM	21	22	26	25	22	79	9LPM
24-09-26	4:15 PM	21	22	26	25	22	75	9LPM
24-09-26	4:30 PM	21	22	26	25	22	72	9LPM
24-09-26	4:45 PM	21	22	26	25	22	74	9LPM
24-09-26	5:00 PM	21	22	26	25	22	73	9LPM

Table 4} CFD RESULTS

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
28-09-26	10:00:00	19	20	23	22	20	75	26 LPM
28-09-26	10:15:00	19	20	23	22	20	75	26 LPM
28-09-26	10:30:00	19	20	23	22	20	76	26 LPM
28-09-26	10:45:00	20	21	24	23	21	76	26 LPM
28-09-26	11:00:00	20	21	24	23	21	76	26 LPM
28-09-26	11:15:00	20	21	24	23	21	75	26 LPM
28-09-26	11:30:00	20	21	24	23	21	78	26 LPM
28-09-26	11:45:00	20	21	24	23	21	78	26 LPM
28-09-26	12:00:00	20	21	24	23	21	79	26 LPM
28-09-26	12:15:00	21	22	25	24	22	81	26 LPM
28-09-26	12:30:00	21	22	25	24	22	83	26 LPM
28-09-26	12:45:00	22	23	26	25	23	82	26 LPM
28-09-26	13:00:00	22	23	26	25	23	83	26 LPM
28-09-26	13:15:00	22	23	26	25	23	85	26 LPM
28-09-26	13:30:00	22	23	26	25	23	85	26 LPM
28-09-26	13:45:00	23	24	27	26	24	85	26 LPM
28-09-26	14:00:00	23	24	27	26	24	83	26 LPM
28-09-26	14:15:00	23	24	27	26	24	82	26 LPM
28-09-26	14:30:00	23	24	27	26	25	82	26 LPM
28-09-26	14:45:00	23	24	27	26	24	82	26 LPM
28-09-26	15:00:00	22	23	26	25	25	81	26 LPM
28-09-26	15:15:00	22	23	26	25	24	80	26 LPM
28-09-26	15:30:00	22	23	26	25	24	79	26 LPM
28-09-26	15:45:00	21	22	25	24	22	79	26 LPM
28-09-26	16:00:00	21	22	25	24	22	76	26 LPM
28-09-26	16:15:00	20	21	24	23	21	75	26 LPM
28-09-26	16:30:00	20	21	24	23	21	74	26 LPM
28-09-26	16:45:00	20	21	24	23	21	74	26 LPM
28-09-26	17:00:00	20	21	24	23	21	74	26 LPM

Table 5} CFD RESULTS

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate

30-09-26	10:00:00	20	23	31	30	21	411	9LPM
30-09-26	10:15:00	20	23	31	30	21	440	9LPM
30-09-26	10:30:00	20	23	31	30	21	445	9LPM
30-09-26	10:45:00	20	23	31	30	21	472	9LPM
30-09-26	11:00:00	21	24	32	31	22	500	9LPM
30-09-26	11:15:00	21	24	32	31	22	540	9LPM
30-09-26	11:30:00	22	25	32	31	23	545	9LPM
30-09-26	11:45:00	22	25	33	32	23	548	9LPM
30-09-26	12:00:00	23	26	34	33	24	550	9LPM
30-09-26	12:15:00	23	26	34	33	24	552	9LPM
30-09-26	12:30:00	23	26	34	33	24	560	9LPM
30-09-26	12:45:00	24	27	35	34	25	575	9LPM
30-09-26	13:00:00	24	27	35	34	25	583	9LPM
30-09-26	13:15:00	25	28	36	35	26	588	9LPM
30-09-26	13:30:00	25	28	37	36	27	596	9LPM
30-09-26	13:45:00	25	28	37	36	26	602	9LPM
30-09-26	14:00:00	25	28	37	36	28	612	9LPM
30-09-26	14:15:00	25	28	37	36	26	625	9LPM
30-09-26	14:30:00	25	28	37	36	27	628	9LPM
30-09-26	14:45:00	24	27	35	34	25	608	9LPM
30-09-26	15:00:00	24	27	35	34	26	598	9LPM
30-09-26	15:15:00	24	27	35	34	25	556	9LPM
30-09-26	15:30:00	24	27	35	34	25	525	9LPM
30-09-26	15:45:00	23	26	34	33	24	480	9LPM
30-09-26	16:00:00	23	26	34	33	24	410	9LPM
30-09-26	16:15:00	23	26	34	33	24	380	9LPM
30-09-26	16:30:00	23	26	34	33	24	298	9LPM
30-09-26	16:45:00	22	25	33	32	23	266	9LPM
30-09-26	17:00:00	22	25	33	32	23	263	9LPM

Table6} CFD RESULTS

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
3/10/2026	10:00:00	21	23	30	30	22	410	26LPM
3/10/2026	10:15:00	21	23	31	30	22	420	26LPM
3/10/2026	10:30:00	21	23	31	30	22	428	26LPM
3/10/2026	10:45:00	21	23	32	31	22	440	26LPM
3/10/2026	11:00:00	22	24	33	32	23	445	26LPM
3/10/2026	11:15:00	22	24	33	32	23	462	26LPM
3/10/2026	11:30:00	22	24	33	32	23	466	26LPM
3/10/2026	11:45:00	23	25	34	33	24	480	26LPM
3/10/2026	12:00:00	23	25	34	33	24	494	26LPM
3/10/2026	12:15:00	24	26	35	34	25	505	26LPM
3/10/2026	12:30:00	24	26	35	34	26	516	26LPM
3/10/2026	12:45:00	24	26	35	34	27	524	26LPM
3/10/2026	13:00:00	24	26	35	34	25	543	26LPM
3/10/2026	13:15:00	25	27	36	35	26	576	26LPM
3/10/2026	13:30:00	25	27	36	35	27	597	26LPM
3/10/2026	13:45:00	25	27	36	35	28	603	26LPM

3/10/2026	14:00:00	26	28	37	36	27	624	26LPM
3/10/2026	14:15:00	26	28	37	35	27	636	26LPM
3/10/2026	14:30:00	25	27	36	35	26	643	26LPM
3/10/2026	14:45:00	25	27	36	34	26	631	26LPM
3/10/2026	15:00:00	24	26	35	34	25	625	26LPM
3/10/2026	15:15:00	24	26	35	34	25	580	26LPM
3/10/2026	15:30:00	24	26	35	34	26	549	26LPM
3/10/2026	15:45:00	24	26	35	34	25	490	26LPM
3/10/2026	16:00:00	23	25	34	33	24	451	26LPM
3/10/2026	16:15:00	23	25	34	33	24	401	26LPM
3/10/2026	16:30:00	23	25	33	32	24	374	26LPM
3/10/2026	16:45:00	23	25	33	32	24	349	26LPM
3/10/2026	17:00:00	22	24	32	31	23	323	26LPM

Table7} CFD RESULTS

Date	Time	T in	T out	T g	Tab	T amb	Radiation	Flow Rate
5/10/2026	10:00:00	19	21	23	22	20	73	9LPM
5/10/2026	10:15:00	19	21	23	22	20	74	9LPM
5/10/2026	10:30:00	19	21	23	22	20	74	9LPM
5/10/2026	10:45:00	20	22	24	23	21	74	9LPM
5/10/2026	11:00:00	20	22	24	23	21	75	9LPM
5/10/2026	11:15:00	20	22	24	23	21	77	9LPM
5/10/2026	11:30:00	20	22	24	23	21	78	9LPM
5/10/2026	11:45:00	20	22	24	23	21	78	9LPM
5/10/2026	12:00:00	21	23	25	24	22	79	9LPM
5/10/2026	12:15:00	21	23	25	24	23	79	9LPM
5/10/2026	12:30:00	21	23	25	24	23	79	9LPM
5/10/2026	12:45:00	21	23	25	24	22	80	9LPM
5/10/2026	13:00:00	22	24	26	25	23	80	9LPM
5/10/2026	13:15:00	22	24	26	25	24	81	9LPM
5/10/2026	13:30:00	23	25	27	26	24	82	9LPM
5/10/2026	13:45:00	23	25	27	26	24	82	9LPM
5/10/2026	14:00:00	24	26	28	27	25	82	9LPM
5/10/2026	14:15:00	24	26	28	27	26	83	9LPM
5/10/2026	14:30:00	25	27	29	28	26	83	9LPM
5/10/2026	14:45:00	25	27	29	28	26	84	9LPM
5/10/2026	15:00:00	24	26	28	27	25	85	9LPM
5/10/2026	15:15:00	24	26	28	27	27	87	9LPM
5/10/2026	15:30:00	23	25	27	26	27	84	9LPM
5/10/2026	15:45:00	23	25	27	26	25	83	9LPM
5/10/2026	16:00:00	23	25	27	26	24	82	9LPM
5/10/2026	16:15:00	22	24	26	25	23	81	9LPM
5/10/2026	16:30:00	22	24	26	25	23	79	9LPM
5/10/2026	16:45:00	21	23	25	24	22	78	9LPM
5/10/2026	17:00:00	21	23	25	24	22	77	9LPM

Table8} CFD RESULTS

6/10/2026	10:00:00	19	20	23	22	20	72	26LPM
6/10/2026	10:15:00	19	20	23	22	20	72	26LPM
6/10/2026	10:30:00	19	20	23	22	20	72	26LPM
6/10/2026	10:45:00	20	21	24	23	21	73	26LPM
6/10/2026	11:00:00	20	21	24	23	21	74	26LPM
6/10/2026	11:15:00	20	21	24	23	21	74	26LPM
6/10/2026	11:30:00	21	22	25	24	22	75	26LPM
6/10/2026	11:45:00	21	22	25	24	22	75	26LPM
6/10/2026	12:00:00	21	22	25	24	22	75	26LPM
6/10/2026	12:15:00	21	22	25	24	22	76	26LPM
6/10/2026	12:30:00	22	23	26	25	23	76	26LPM
6/10/2026	12:45:00	22	23	26	25	23	77	26LPM
6/10/2026	13:00:00	22	23	26	25	23	78	26LPM
6/10/2026	13:15:00	22	23	26	25	23	79	26LPM
6/10/2026	13:30:00	23	24	27	26	24	81	26LPM
6/10/2026	13:45:00	23	24	27	26	24	82	26LPM
6/10/2026	14:00:00	23	24	27	26	24	82	26LPM
6/10/2026	14:15:00	23	24	27	26	24	83	26LPM
6/10/2026	14:30:00	24	25	28	27	26	83	26LPM
6/10/2026	14:45:00	24	25	28	27	26	84	26LPM
6/10/2026	15:00:00	24	25	28	27	25	85	26LPM
6/10/2026	15:15:00	23	24	27	26	24	84	26LPM
6/10/2026	15:30:00	23	24	27	26	26	83	26LPM
6/10/2026	15:45:00	23	24	27	26	24	82	26LPM
6/10/2026	16:00:00	22	23	26	25	23	82	26LPM
6/10/2026	16:15:00	21	22	25	24	22	80	26LPM
6/10/2026	16:30:00	21	22	25	24	22	79	26LPM
6/10/2026	16:45:00	20	21	24	23	21	78	26LPM
6/10/2026	17:00:00	20	21	24	23	21	77	26LPM

CHAPTER 5

THEORETICAL ANALYSIS AND CALCULATIONS

5.1 Introduction

The theoretical analysis of a Solar Parabolic Trough Collector is necessary to evaluate its thermal performance. The experimental readings obtained in Chapter 4 are used to calculate the mass flow rate, useful heat gain, solar energy input and thermal efficiency of the collector.

In this project, the calculations are performed for normal water and saline water at flow rates of 9 LPM and 26 LPM under full and half solar radiation conditions. The calculated values are then compared to understand how the working fluid, flow rate and solar radiation affect the performance of the collector.

5.2 Calculation of Solar Energy Input

The incident solar power is calculated using:

$$Qs = Aa * G$$

Where:

- Qs = incident solar power (W)
- Aa = aperture area of the reflector (m^2)
- G = average measured solar radiation (W/m^2)

9 LPM Full Radiation

$$Qs = Aa \times 514.3$$

Thus,

$$Qs = 514.3\ Aa\ W$$

5.3 Test calculations

condition	Tin avg. (°C)	Average outlet temp. (°C)	Average radiation (W/m²)
Normal water, 9 LPM, Full	**24.17**	**26.17**	**514.3**
Normal water, 26 LPM, Full	**21.72**	**22.28**	**448.2**
NW, 9 LPM, Low	**21.41**	**21.55**	**82.3**
Normal water, 26 LPM, Low	**21.03**	**21.03**	**79.1**
Saline water, 9 LPM, Full	**22.93**	**24.07**	**508.8**
Saline water, 26 LPM, Full	**23.41**	**24.21**	**502.9**
Saline water, 9 LPM, Low	**21.79**	**21.97**	**79.8**
Saline water, 26 LPM, Low	**21.59**	**21.66**	**78.4**

5.4 Calculation of Mass Flow Rate

The mass flow rate is calculated using:

$M = \rho * V$

For water, approximate density is taken as 1000 kg/m^3

$\dot{V} =$ Flow rate (LPM) / 60000

For 9 LPM:

$$\dot{m} = 1000 \times 9 / 60000$$
$$\dot{m} = 0.15 \text{ kg/s}$$

For 26 LPM:

$$\dot{m} = 1000 \times 26 / 60000$$
$$\dot{m} = 0.4333 \text{ kg/s}$$

In saline water due to change in density can may change the mass flow rate

5.4 Calculation of Useful Heat Gain

Useful heat gain is calculated using:

$$Qu = \dot{m} Cp (Tout - Tin)$$

For an initial comparison, the specific heat capacity of water is taken as approximately 4180 J/(kg·K). The same value is provisionally used for saline water because the salt concentration has not been provided. Therefore, the saline-water results are estimates, not exact property-based calculations.

Example 1: Normal water at 9 LPM under full radiation

Given:

- Mass flow rate = 0.15kg/s
- Specific heat capacity = 4180J/(kg·K)
- Average temperature difference = 2 degree Celcius

Substituting:

Qu = 0.15 × 4180 × 2
Qu = 1254 W

Therefore, the estimated useful heat gain is **1.254 kW.**

Example 2: Normal water at 26 LPM under full radiation

Qu = 0.4333 × 4180 × 0.552
Qu ≈ 999.4 W

5.5 Aperture Area of the Collector

Given:

- Aperture width, Wa = 1250 mm = 1.25 m
- Collector length, Lc = 3m (approximately)

Formula:

Aa = Wa × Lc
Aa = 1.25 × 3
Aa = 3.75 m^2

Therefore, the approximate aperture area of the Solar Parabolic Trough Collector is 3.75 m².

5.6 Solar Energy Input

The incident solar power is calculated using:

Qs = Aa × G

For normal water at 9 LPM under full radiation, using the average measured radiation of 514.3 W/m^2

Qs = 3.75× 514.3
Qs = 1928.63 W

5.7 Calculation of Thermal Efficiency

Thermal efficiency is the ratio of useful heat gained by the working fluid to the incident solar power received by the collector.

ηth = Qu Qs × 100

For normal water at 9 LPM under full radiation:

- **Useful heat gain, Qu = 1254W**
- **Solar input, Qs = 1928.63W**

ηth = 12541928.63 × 100
ηth ≈ 65.02%

Using the average readings and the provisional heat-gain calculations from the estimated results for all eight conditions are:

Working fluid	Flow rate	Radiation	Solar input (W)	Useful heat gain (W)	Estimated efficiency
Normal water	**9 LPM**	**Full**	**1928.63**	**1254.0**	**65.02%**
Normal water	**26 LPM**	**Full**	**1680.75**	**999.4**	**59.46%**
Normal water	**9 LPM**	**Low**	**308.63**	**86.5**	**28.03%**
Normal water	**26 LPM**	**Low**	**296.63**	**0.0**	**0.00%**
Saline water	**9 LPM**	**Full**	**1908.00**	**713.5**	**37.40%**
Saline water	**26 LPM**	**Full**	**1885.88**	**1436.6**	**76.20%**
Saline water	**9 LPM**	**Low**	**299.25**	**108.1**	**36.12%**
Saline water	**26 LPM**	**Low**	**294.00**	**124.9**	**42.48%**

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Experimental Observations

The experimental investigation was carried out to evaluate the thermal performance of the fabricated Solar Parabolic Trough Collector (PTC). The effect of working fluid, flow rate and solar radiation on the temperature rise and useful heat gain was studied using normal water and saline water.

Two volumetric flow rates, namely 9 LPM and 26 LPM, were considered. Experiments were conducted under full and low solar radiation conditions. The inlet temperature, outlet temperature and average solar radiation were used to compare the performance of the collector under different operating conditions.

The temperature rise of the working fluid was determined using the following equation:

Where:

- (T) = Temperature rise of the working fluid (°C)
- (Tout) = Average outlet temperature (°C)

Table 6.1: Summary of Experimental Observations

Working fluid	Flow rate (LPM)	Radiation condition	Inlet temp. (°C)	Outlet temp. (°C)	Radiation (W/m^2)	Temperature rise (°C)
Normal water	9	Full	24.17	26.17	514.3	2.00
Normal water	26	Full	21.72	22.28	448.2	0.56
Normal water	9	Low	21.41	21.55	82.3	0.14
Normal water	26	Low	21.03	21.03	79.1	0.00
Saline water	9	Full	22.93	24.07	508.8	1.14
Saline water	26	Full	23.41	24.21	502.9	0.80
Saline water	9	Low	21.79	21.97	79.8	0.18
Saline water	26	Low	21.59	21.66	78.4	0.07

It is found that the temperature rise is dependent on the operating conditions from the observations. For normal water, the highest temperature rise was recorded in the full radiation condition as illustrated in the table. Also measured and found to be 0.5 °C under full radiation for saline water at 26 LPM.

The temperature differences under low radiation were relatively small. The temperature rise was reported with the same precision for normal water at 26 LPM. This means that the temperatures at the inlet and outlet were the same (rounded to two decimal places).

The outcomes show that flow rate variable and solar radiation are important parameters for the study of the thermal performance of the collector. The temperature rise, however, is not enough to decide on the optimum operating condition. Thermal efficiency and heat gain are also important factors to consider.

6.2 Normal Water at 9 LPM

The performance of Parabolic Trough solar collector was investigated for normal water at 9 litres per minute (LPM) flow rate for full solar radiation and low solar radiation conditions.

6.2.1 Full Radiation Condition

The average inlet temperature was 24.17°C and the average outlet temperature was 26.17°C. The average temperature rise that was measured was 2.00°C. The average solar radiations were 514.3W/m².

The results show that a transfer of thermal energy took place from the collector to the normal water in the experiment. The temperature rise that was observed indicates that the receiver absorbed concentrated solar radiation and radiated a part of the absorbed heat to the flowing water.

At lower flow rates (26 LPM), the water stayed in the receiver longer than at higher flow rates (at similar flow paths and operating conditions), which may have been helpful for separation.A relatively low flow rate allowed the water to stay in the receiver for a longer time than at higher flow rates (assuming similar flow paths and operating conditions) which could have been beneficial in separation. This could have accounted for the greater increase in water temperature on a mass basis.

6.2.2 Low Radiation Condition

In low radiation, inlet and outlet temperature were 21.41°C and 21.55°C respectively, on average. The temperature increase was therefore 0.14°C and there was an average of 82.3 W/m² solar radiation.

The rise in temperature was much less than under full radiation. Due to reduced availability of solar energy, the transfer of heat to the working fluid was also reduced.

6.2.3 Result Analysis

A comparison of both conditions show, that normal water at 9 LPM had higher temperature rise under the condition of full radiation as under low radiation condition.

When the test condition was reduced from full to low radiation the temperature rise was reduced from 2.00°C to 0.14°C. This equates to about a 93% decrease in the measured rise in temperature.

The comparison helps in the observation that the temperature response of the collector is greatly influenced by solar radiation. However, to determine the actual thermal efficiency, the collector aperture area and an accurate estimation of incident solar power are needed.

Saline Water 6.3 LPM

The saline water was tested at 9LPM to study the thermal behaviour of the saline water in the solar parabolic trough collector.

6.3.1 Full Radiation Condition

The mean inlet and outlet temperature of saline water for full radiation was 22.93 and 24.07°C respectively. The temperature rise observed was 1.14 °C. The average of measured solar radiation was 508.8 W/m2.

This temperature difference is due to the transfer of heat to the saline water as it flows through the receiver and is an indicator that the receiver was transferring heat to the saline water inlet.

The temperature rise was less than in the case of normal water at the same flow rate, with full radiation. But there were differences in the radiation level measured and initial temperature in the two tests. This difference should not, therefore, solely be blamed on the kind of working fluid.

6.3.2 Low Radiation Condition

The inlet temperature was 21.79°C and outlet was 21.97°C under low radiation. The temperature rise was 0.18°C and the average solar radiation was 79.8 W/m².

The temperature rise observed was small which is in line with the decreased availability of solar energy during the test.

6.3.3 Result Analysis

The temperature elevation of the saline water reduced from 1.14°C in the case of full radiation to 0.18°C in the case of low radiation. This is about 84.2% less.

The results show that a measurable amount of heat was delivered to saline water in both the two radiation conditions tested. However, the recorded temperature elevation does not prove if saline water is more suitable as a working fluid as compared to normal water.

All the following factors must be taken into account to make a reliable comparison: fluid density, specific heat capacity, salt concentration, mass flow rate, and the actual experimental conditions.

6.4 Normal Water @ 26 LPM

The collector was also tested with normal water having a higher flow rate of 26 LPM. The objective is to find the difference between low flow rate and high flow rate

6.4.1 Full Radiation Condition

In full radiation, the inlet and outlet temperature was 21.72°C and 22.28°C. The increase in temperature was 0.56°C and the average solar radiation measured was 448.2 W/m².

The temperature rise was found to be less than that at 9 LPM which was 2.00°C. This could be due to the increased flow rate bringing more water through the receiver in the given amount of time, spreading out the amount of thermal energy within more fluids.

But a lesser temperature rise does not necessarily mean less useful heat gain. Useful heat gain rate is a function of the mass flow rate and the temperature rise.

6.4.2 Low Radiation Condition

However, as the radiation was low, the average inlet temperature was 21.03°C and the average outlet temperature was also 21.03°C, making the temperature rise from the given averages zero. The average solar radiation was 79.1 W/m².

When the temperature difference is not measurable, there was no temperature increase that could be measured at the level of the reported differences. Does not imply that no heat was transferred, as small temperature changes may be within the instrument resolution or experimental error.

6.4.3 Result Analysis

The temperature rise for a 26LPM at full radiation was 0.56°C and at low radiation was 0.00°C.

The conclusions indicate that the collector's capability of raising the temperature to a measurable level was decreased with low radiation. A lower temperature rise, also, was observed at higher flow rate compared to 9 LPM under full radiation.

Hence, the choice of operating flow rate must be based on the overall consideration of the temperature rise, useful heat gain, pumping demand and thermal efficiency instead of outlet temperature only.

6.5 Saline Water at 26 LPM

The saline water was tested at 26 LPM to find its thermal response at higher flow rate.

6.5.1 Full Radiation Condition

An average temperature of 23.41°C was measured at the inlet and an average temperature of 24.21°C was measured at the outlet under full radiation. The recorded temperature rise was 0.80 °C. The mean solar radiation at that time was 502.9W/m².

The results show that the temperature of the saline water changed as it flowed through the receiver tube.

The temperature rise was less in the saline water at 26 LPM than that at 9 LPM. When

considering the amount of heat energy carried by the fluid per unit time, however, the higher mass flow rate may result in a greater amount of heat.

6.5.2 Low Radiation Condition

Under low radiation, the average inlet and outlet temperatures were 21.59°C and 21.66°C, respectively. The temperature rise was 0.07°C, with an average solar radiation of 78.4 W/m².

The small temperature difference indicates limited heating of the saline water under the low-radiation condition.

6.5.3 Result Analysis

The temperature rise decreased from 0.80°C under full radiation to 0.07°C under low radiation, representing a reduction of approximately 91.3%.

The measurements show that full radiation produced a larger temperature rise than low radiation for saline water at 26 LPM.

The performance of saline water at the higher flow rate should also be assessed through useful heat gain calculations. Since the salt concentration was not specified in the available calculation data, the specific heat capacity used for saline water must be treated as an approximation.

6.6 Effect of Full and Low Radiation

Solar radiation is one of the major factors affecting the performance of a parabolic trough collector.
.

Table 6.2: Comparison of Temperature Rise Under Full and Low Radiation

Working fluid	Flow rate (LPM)	Full-radiation temperature rise (°C)	Low-radiation temperature rise (°C)
Normal water	9	2.00	0.14
Normal water	26	0.56	0.00
Saline water	9	1.14	0.18
Saline water	26	0.80	0.07

The table shows that the temperature rise was greater under full radiation for all four combinations of working fluid and flow rate.

The low-radiation measurements ranged from 78.4 to 82.3 W/m², while the full-radiation measurements ranged from 448.2 to 514.3 W/m².

6.7 Comparison of Working Fluids

Table 6.3: Comparison of Normal Water and Saline Water

Radiation condition	Flow rate (LPM)	Normal water ΔT (°C)	Saline water ΔT (°C)	Higher measured temperature rise
Full	9	2.00	1.14	Normal water
Full	26	0.56	0.80	Saline water
Low	9	0.14	0.18	Saline water
Low	26	0.00	0.07	Saline water

The temperature rise of normal water was higher than saline water at 9 LPM full radiation.

The temperature rise of saline water was more at 26 LPM at full radiation.

In low radiation, saline water had slightly higher temperature rises
at both flow rates. The differences with low radiation, however, were small at 26 LPM.

It is observed that the ratio of temperature response of the two fluids varied with the operating condition. The measurements available are not a clear indicator of superiority for either fluid.

However, the comparison should be taken with a pinch of salt since the inlet temperature and measured solar radiation were not the same for each test pair. Further, there was no information on saline-water concentration and related thermal properties in the calculation summary available.

If it is to be compared with an accurate accuracy, then the mass flow rate and specific heat capacity of each fluid should be calculated. Then the useful heat gain and the thermal efficiency of each test should be calculated.

6.8 Graphical Analysis

Graphs give a clear representation of the experimental observations and can be used to determine the influence of the operating parameters on the collector's thermal response. The following graphs are suggested for this project.

Graph of temperature rise at a flow rate of 9 lpm.

The bar graph should be plotted as follows: working fluid and radiation condition should be plotted on the horizontal axis and temperature rise should be plotted on the vertical axis.

The values to be plotted are:

Test condition	**Temperature rise (°C)**
Normal water, full radiation	2.00
Normal water, low radiation	0.14
Saline water, full radiation	1.14
Saline water, low radiation	0.18

The graph will show that normal water at 9 LPM recorded the highest temperature rise among these four conditions.

6.8.3 Graph of Solar Radiation and Temperature Rise

A comparative graph can be prepared with average solar radiation on the horizontal axis and temperature rise on the vertical axis. Plot the eight experimental points from Table 6.1 separately, using different symbols or labels for each fluid and flow rate.

The graph can then be used to examine the relationship between measured solar radiation and temperature rise. Do not force the points into a straight line: other variables also affect the results, including flow rate, inlet temperature, and heat losses.

6.8.4 Graph of Estimated Useful Heat Gain

Useful heat gain can calculate on flow rate, specific heat :

The number of approximately is 4180 for project for specific heat capacity of water.
Saline water.
The average temperature is in table for the following approximation to be calculate in the different temperature extremes.

Fluid	**(LPM)**	**Radiation**	**heat gain (W)**
water	9	Full	1254
water	26	Full	1014

water	9	Low	88
water	26	Low	0
Saline water	9	Full	716
Saline water	26	Full	1449
Saline water	9	Low	113
Saline water	26	Low	127

The values are calculated are approximate.

6.9 Discussion of Results

The experimental investigation was carried out to analyze the effect of fluid and solar radiation of fabricated solar collector.

6.9.1 Influence of Solar Radiation

The following observations influence by solar radiation is given by 6.91 temperature rise was more in full radiation than low radiation .

6.9.2 Influence of Flow Rate

It is found that the results are related with increasing flow; generally decrease in temperature rise per. This shows that rise in temperature reduces from 2 degree celcius to 0.56 degrees Celsius.
The change in temperature does not mean decrease in usefull heat gain as the mass flow of the water in the system increase.

6.9.3 Influence of Working Fluid

The result of comparing water and saline water are different in different conditions the higher the temperature rise the higher temperature was achieved.

6.9.5 Limitations Affecting the Interpretation

When interpreting the experimental results the following points should be taken into account: In the first place there was a variation in solar radiation from test to test, even when the tests were

1. Radiation under a similar general radiation category.
2. The inlet temperature was different for the test conditions.
3. The average temperatures reported are rounded and may have a margin of error of 0.00°C can't be interpreted as no heat transfer.
4. Concentrations of saline water and hence its specific heat capacity were not present in the calculation summary.
5. The verified collector aperture area and the verified reference value of the collector are needed for a final thermal-efficiency comparison.
6. Useful heat-gain calculations that are reliable.

CFD analysis.

6.10 CFD ANALYSIS OF SOLAR PARABOLIC TROUGH COLLECTOR

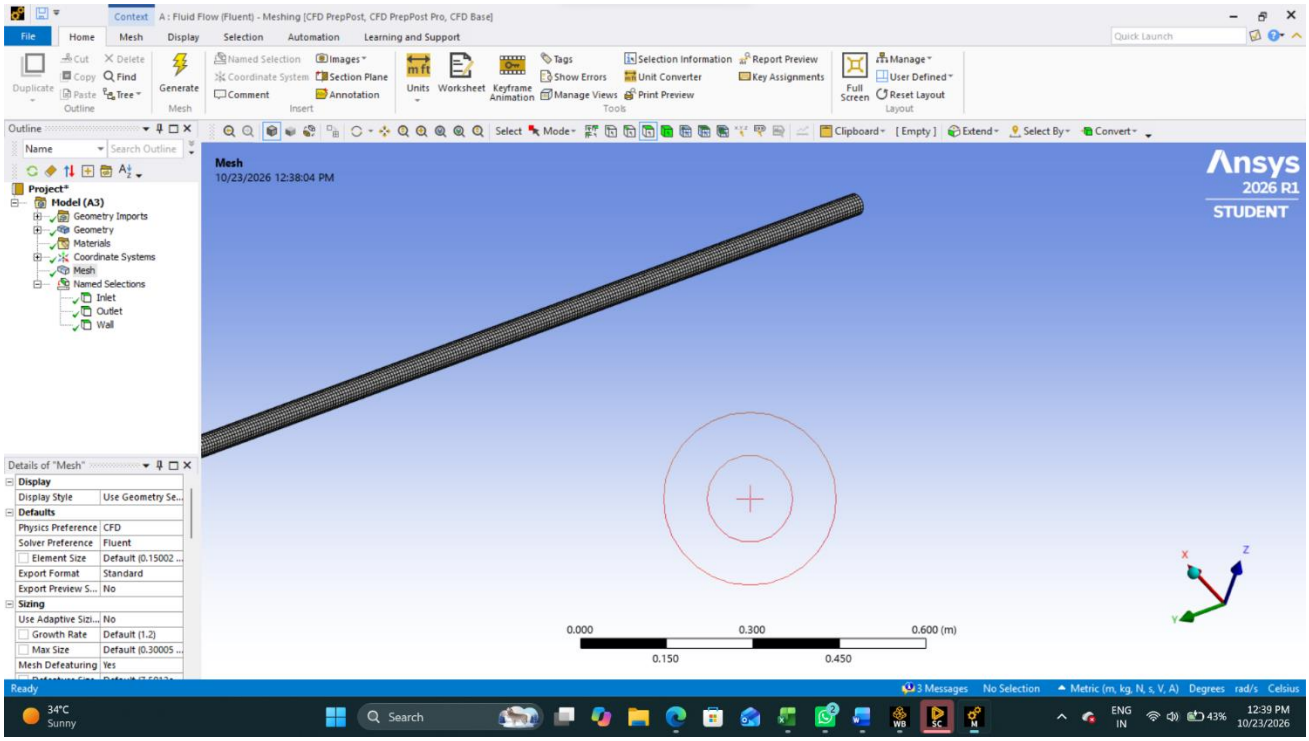
6.10.1 Introduction

Computational Fluid Dynamics (CFD) analysis was carried out using ANSYS Fluent 2026 R1 software. The computational Fluid Dynamics (CFD) analysis was done using the ANSYS Fluent 2026 R1 software. The Solar receiver tube was constructed, and the flow and temperature distribution within the tube was investigated.
The numerical study was carried out to supplement the experimental investigation and understanding of the thermal behaviour of the working fluid inside the receiver tube.
The experiments were carried out with normal water with flowrate of 9 LPM under full solar radiation. The experimentally measured outlet temperature for this condition was reported as 27°C. The temperature distribution was simulated using the CFD model, and the results were represented graphically.

6.10.2 Geometry and Mesh Generation

Geometry of the receiver tube was created in the ANSYS Meshing software. A computational mesh was the fluid domain is subdivided into smaller elements called "finite elements" that are used in a numerical analysis. The inlet, named selections were used to determine the boundaries of outlet and wall.

The mesh will be generated which will give the computational domain for the solution of the governing equations of fluid flow and heat transfer. The mesh has an appropriate resolution and quality.

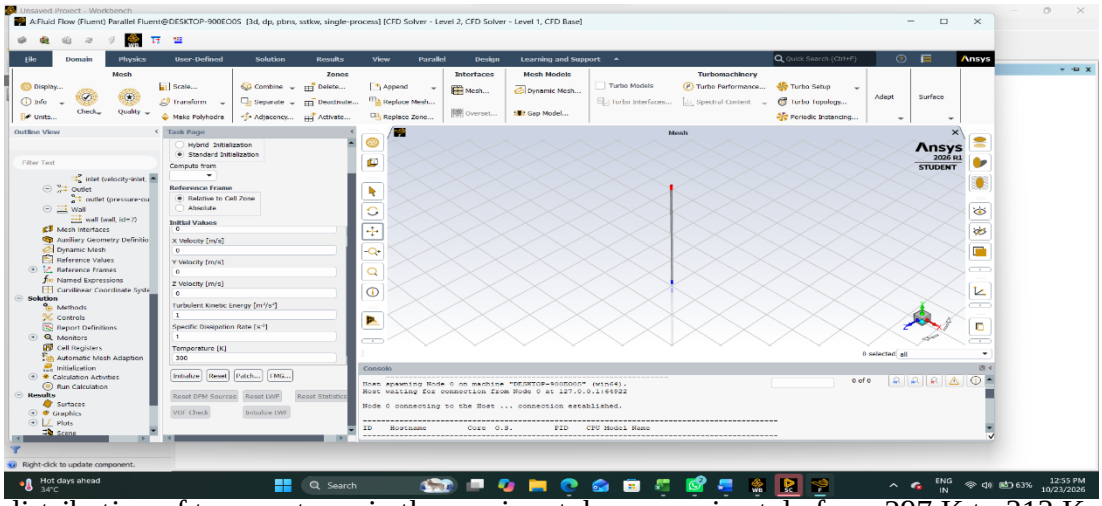


6.10.3 CFD Setup and Numerical Calculation

The receiver tube with a meshed surface was analysed in ANSYS Fluent 2026 R1 software the model included the inlet, outlet and wall boundary zones. The solver had been initialized prior to the following operation:
numerical calculation.
The initialization panel showed an initial temperature to be 300 K, which is equal to approximately 26.85°C. The Fluent interface had a calculation set up for 50 iterations and the console was output showed the message “Calculation complete.” However, the available the solution is not necessarily fully converged, as indicated by screenshots. To make a valid comparison with the experiment, the CFD setup needs to be made as similar to the experiment as possible. The inlet temperature, inlet flow rate, working fluid properties and thermal boundary are experimental variables. As near to the actual condition as possible as closely as possible.

6.10.4 Static Temperature Contour Analysis

The static temperature contour from ANSYS Fluent shows the ANSYS calculated



distribution of temperatures in the receiver tube approximately from 297 K to 313 K.

The contour image is predominantly in the blue color, and there are areas of higher contour with a darker blue. The temperature at approximately the end of the tube viewed. These variations are the computed spatial the temperature distribution in the computed domain.

The highest temperature represented on the contour is around 313K. This value is the highest value of the temperature field displayed and cannot be used directly as . the outlet temperature. The outlet boundary is used to calculate the outlet temperature. Appropriate average-temperature calculation used.

6.10.5 Velocity Magnitude Contour Analysis

The magnitude of the velocity contour shows the velocity distribution within the receiver tube. The velocity legend shown is between about 0 and 0.192 m/s. The increased velocity magnitudes can be seen in the central portion of the displayed section, and lower values are found in the vicinity of the surrounding wall.

This distribution is a qualitative view of the flow properties in the simulated area. The peak velocity shown is about 0.192 m/s. This value needs to be verified. For the specific inlet flow rate.

6.10.6 Experimental and CFD Results Comparison

The experimental outlet water temperature of normal water at 9LPM in the case of full solar radiation was reported as 27°C. The outlet temperature of the CFD needs to be obtained from ANSYS Fluent and so does the inlet temperature.

Calculate the difference in numbers between the experimental and simulated results.
The difference in temperature may be determined by:

Temperature difference = CFD outlet temperature − Experimental outlet temperature
When operating condition:
Experimental outlet temp = 27 °C

Therefore:
Temperature difference is equal to CFD outlet temperature - 27°C.
A positive result means that the temperature of the CFD outlet is greater than the experimental value.

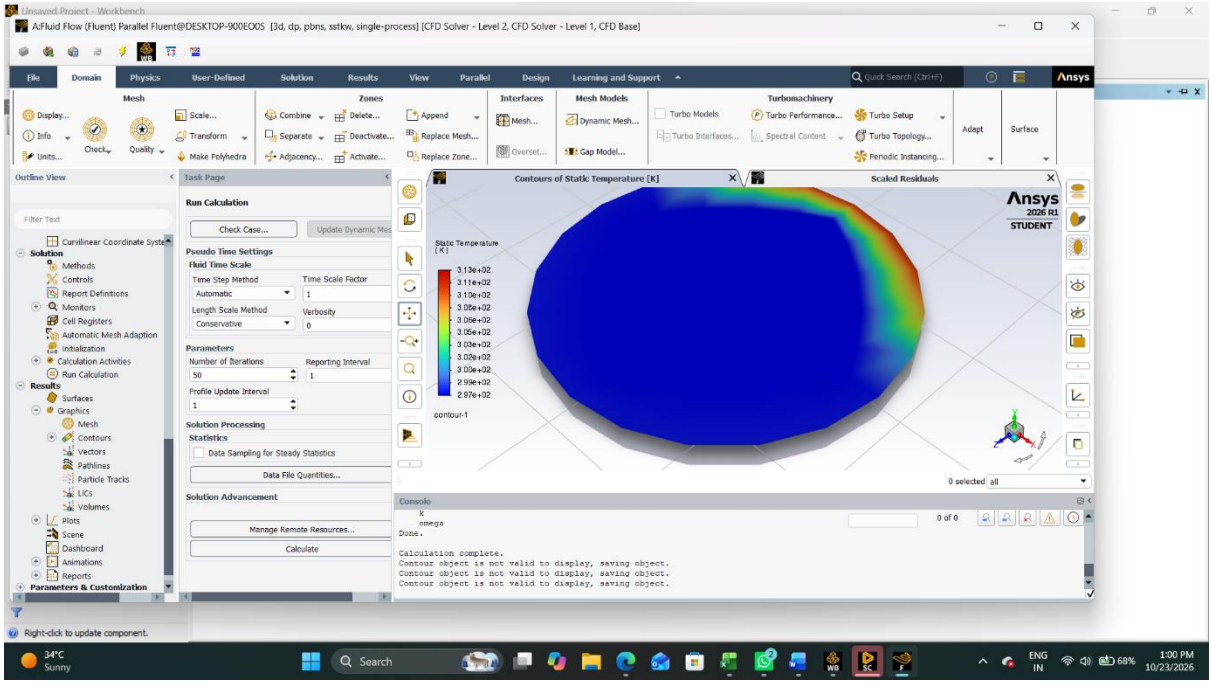
Table 6.4: Experimental and CFD Comparison for Normal Water at 9 LPM Under Full

Radiation

Parameter	Experimental result	CFD result
Working fluid	Normal water	Confirmed from model settings
Flow rate	9 LPM	Confirmed from inlet settings
Radiation	Full radiation	Boundary condition verified
T out	27°C	Extracted
T difference	Reference	27°C
Maximum static temperature	NA	Approximately 313 K
Maximum velocity	NA	0.192 m/s

6.10.7 Discussion of CFD Results

The CFD analysis gives a graphical representation of the temperature and velocity distributions inside the receiver tube. The temperature contour displays the spatial variations of the computed thermal field and the velocity contour depicts the distribution of fluid velocity in the shown section.

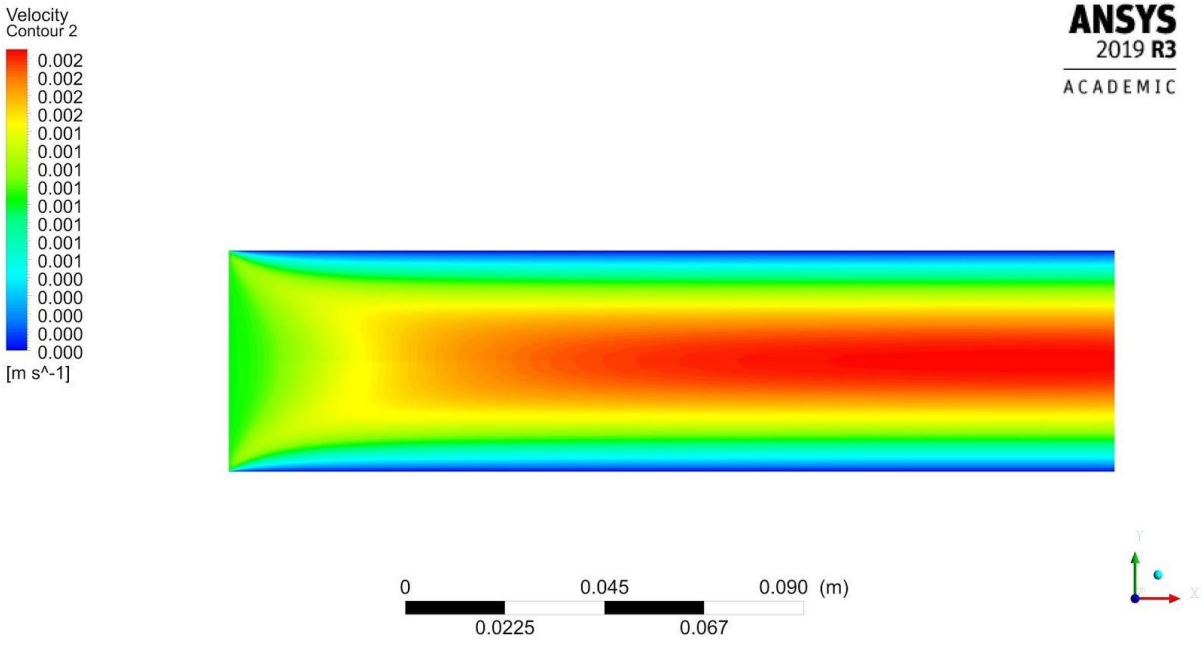


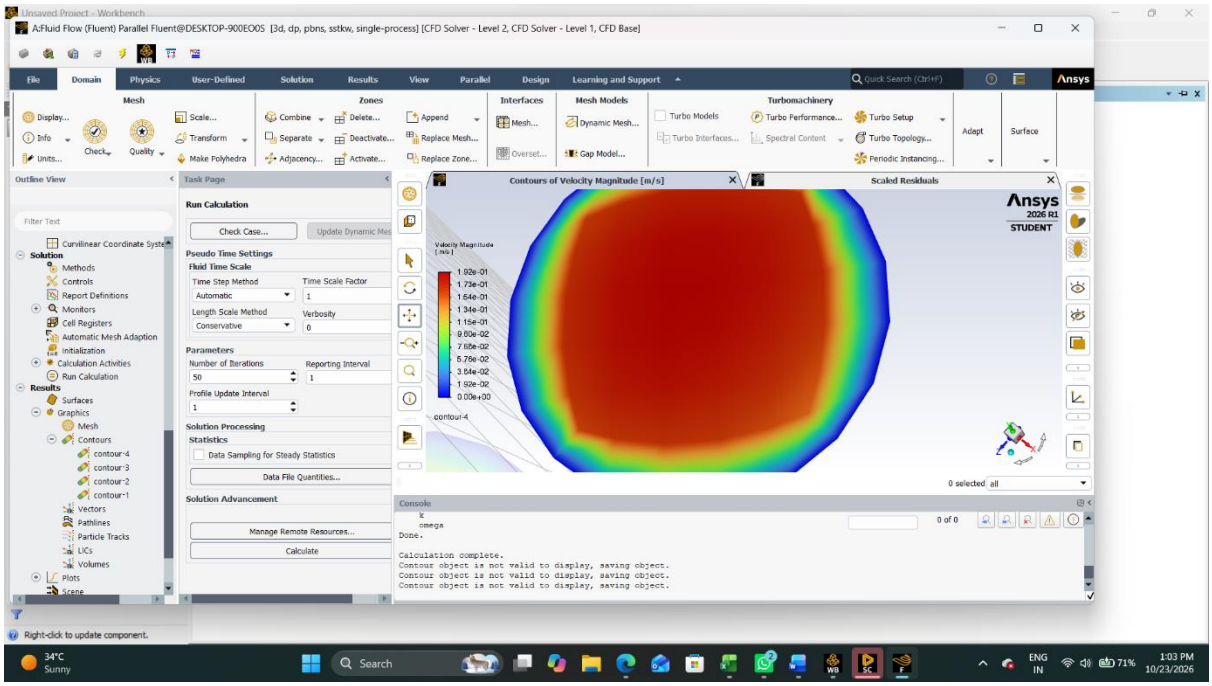
The experimental study indicated the outlet temperature for normal water at 9 LPM under full radiation was 27°C. The screenshots of the CFD provided do not provide a validated numerical value of the average outlet temperature. So the exact difference of experimental and CFD results cannot be established yet.

The difference between experiment and cfd results are due to the heat loss also measurement inaccuracy and surrounding

The residual history and relevant boundary conditions should also be checked before drawing conclusions about the accuracy of the numerical model.

Overall, the CFD analysis supplements the experimental investigation by providing information about the temperature and velocity fields inside the receiver tube. Further quantitative analysis is required to determine the agreement between the experimental and numerical results and to establish the thermal performance of the collector.




Unsaved Project - Workbench
A:Fluid Flow (Fluent) Parallel Fluent@DESKTOP-900EO0S [3d, dp, pbns, sstkw, single-process] [CFD Solver - Level 2, CFD Solver - Level 1, CFD Base]
File
Domain
Physics
User-Defined
Solution
Results
View
Parallel
Design
Learning and Support
Ansys
Mesh
Zones
Interfaces
Mesh Models
Turbomachinery
Adapt
Surface
Outline View
Task Page
Run Calculation
Check Case...
Pseudo Time Settings
Fluid Time Scale
Time Step Method
Automatic
Time Scale Factor
Length Scale Method
Conservative
Verbosity
Parameters
Number of Iterations
Reporting Interval
Profile Update Interval
Solution Processing
Statistics
Data Sampling for Steady Statistics
Data File Quantities...
Solution Advancement
Manage Remote Resources...
Calculate
Contours of Velocity Magnitude [m/s]
Scaled Residuals
Ansys
2026 R1
STUDENT
Velocity Magnitude [m/s]
1.92e-01
1.73e-01
1.54e-01
1.34e-01
1.15e-01
9.60e-02
7.68e-02
5.76e-02
3.84e-02
1.92e-02
0.00e+00
contour-4
0 selected
all
Console
k
omega
Done.
Calculation complete.
Contour object is not valid to display, saving object.
Contour object is not valid to display, saving object.
Contour object is not valid to display, saving object.
0 of 0
Right-click to update component.
34°C
Sunny
Search
ENG
IN
71%
1:03 PM
10/23/2026

CHAPTER 7
OPTIMIZATION AND PERFORMANCE EVALUATION

7.1 Introduction

Solar PTC performance depends upon solar radiation, water flow rate, fluid properties, and heat transfer conditions among many others factors. This project tested the performance of the collector with normal water and saline water flowing through the collector at two flow rates which are 9 LPM and 26 LPM under full solar radiation and low solar radiation conditions.
The goal is to compare the experimental data and decide on which operating conditions worked best with recommendations to improve thermal performance.

7.2 Performance Comparison

Fluid	Flow Rate	Radiation	Inlet Temp. (°C)	Outlet Temp. (°C)	Temperature Rise (°C)
Normal water	9 LPM	Full	24.17	26.17	2.00
Normal water	26 LPM	Full	21.72	22.28	0.56
Normal water	9 LPM	Low	21.41	21.55	0.14
Normal water	26 LPM	Low	21.03	21.03	0.00
Saline water	9 LPM	Full	22.93	24.07	1.14
Saline water	26 LPM	Full	23.41	24.21	0.80
Saline water	9 LPM	Low	21.79	21.97	0.18
Saline water	26 LPM	Low	21.59	21.66	0.07

The data indicates that full solar radiation created a higher temperature rise for the majority of the tests when compared with low radiation. Maximum recorded temperature rise of normal water was 2.00° C at 9 LPM. Maximum recorded temperature rise of saline water was 1.14° C at 9 LPM. Temperature rise occured at full radiation during both of these tests.

At 26 LPM the temperature rise was lower for both fluids being tested during full radiation tests. Perhaps the fluid does not spend enough time in the collector to gain as much heat. Total useful heat gain would also be affected by mass flow rate and specific heat capacity of the fluid.

This data should be viewed along with the solar radiation and inlet temperature values of each test as identical conditions were not maintained throughout all tests.

7.3 Identification of Better Operating Conditions

From observations of temperature measurements during the experiment, certain operating conditions were chosen:

1. Full Solar Radiation: Tests with high solar radiation produced a larger change in temperature than those with low radiation.
2. Normal Water 9 LPM: Yielded the largest recorded temperature rise of all data from the experiment at 2.00°C.
3. Saline Water 9 LPM: Showed a temperature rise of 1.14°C at full radiation.
4. Higher Flow Rate: Showed a lower rise in temperature at 26 LPM in recorded full-radiation tests. The higher mass flow would produce more gallons of heated fluid per time interval.
5. Very low radiation: in low radiation the temperature rise is very low

These conditions combine to show that normal water at 9 LPM with full solar radiation gave the largest measured temperature rise at the outlet.

7.4 Factors Affecting Collector Performance

1.Solar radiation affects the performance. In low radiation heat gain is less and and in high radiation is high
2.Flow rate affects the performance of ptc
3.Density of fluid, viscosity and specific heat also plays very important role
4.There are various heat and optical losses are include

7.5 Recommendations for Improvement

1.Provide strong support structure and alignment to setup
2.reduce heat loss by proper thermal insulation
3.align reflector like so they can maximize the concentration
4.calibration of measuring instruments
5.smooth flow of working fluid
6.optimize the flow rate

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

We done this project to analyze the performance of ptc in different scenarios
We did 8 types of experiments in low and high radiation, in different flow rates and with 2 types of fluids
We analyze that in normal water at 9 lpm and in high radiation there is the temperature increase by 2 degree Celsius
In saline water at 9 lpm the average increase in temperature is 1.14
It shows that higher the flow rate does not mean that high energy gain
For validated the results we did cfd
We use ansys fluent to validate the results
We use different kind of contours like meshing, static temperature, velocity magnitude to find the various parameters
Cfd does not consider most of losses

8.2 Limitations of the Project

Several limitations were faced, which affected the results of the project:
Energy available for heating varied during testing, depending on weather conditions (solar radiation).
Low flow rates: We considered only two flow rates, 9 LPM and 26 LPM.
Reduction of radiation conditions: only full radiation and low radiation conditions were studied and excluded a wider range of radiation conditions with controlled solar inputs.
Measurement errors: Slight inaccuracies in the measurement of temperature, flow rate and sun radiation may have affected the results.
Properties of saline water: It is important to know the salinity concentration in order to correctly calculate other properties such as specific heat capacity.
Heat losses: Heat loss from the receiver tube and pipes may have reduced the heat transfer to the fluid.
Limited testing time: The results are based on the specific testing conditions, and cannot be used to predict performance for a whole day or throughout the seasons.
The following are required to have a reliable quantitative comparison between CFD and experimental results: Consistent boundary conditions, Appropriate fluid properties, Verified outlet-average temperatures

8.3 Future Scope

Improvements and investigations that we can explore in our future work:
Experiment with other flow rates: Additional tests can be performed at different flow rates to determine the best flow rate to operate at for the greatest heat gain and thermal efficiency.
Investigate various saline solutions: Testing saline solutions of varying concentrations will help us determine how these affect heating of the fluids and heat transfer.
Receiver design : Investigate selective absorber coatings, receiver insulation or improved

receiver configuration to reduce heat losses.
Implementing solar tracking: Implementing a tracking mechanism may be useful in maintaining the perfect alignment of the reflector with the sun to achieve better results.
Improve CFD Analysis: Using a finer mesh, radiation modelling and validated boundary conditions will help the numerical results be more accurate.
Validate experimental and CFD results: Under the same operating condition it is very important to validate the results from the experiments with the results from the CFD simulation.
Test long term performance: Tests should be conducted over different seasons and weather conditions to gain a better understanding of the collector's long term performance.
Practical applications: The effectiveness of the collector can be evaluated for domestic hot water, agricultural processing and low temperature industrial heating.

REFERENCES:-

Add the exact books, research papers, and technical documents consulted during the project. The following are suggested foundational references; verify the edition and publication details of each source before submitting the report.

1. Duffie, J. A., and Beckman, W. A., *Solar Engineering of Thermal Processes*, Wiley.
2. Kalogirou, S. A., *Solar Energy Engineering: Processes and Systems*, Academic Press.
3. ANSYS, *ANSYS Fluent User's Guide*, documentation corresponding to the software version used.
4. Multiobjective design optimization of parabolic trough collectors
5. Applicable standards, laboratory manuals, and equipment manuals like rotameter thermocouple sensor

APPENDIX

Fluid	**Flow Rate**	**Radiation Condition**	**Average Inlet Temperature (°C)**	**Average Outlet Temperature (°C)**	**Temperature Rise (°C)**	**Average Solar Radiation (W/m²)**
Normal water	9 LPM	Full	24.17	26.17	2.00	514.3
Normal water	26 LPM	Full	21.72	22.28	0.56	448.2
Normal water	9 LPM	Low	21.41	21.55	0.14	82.3
Normal water	26 LPM	Low	21.03	21.03	0.00	79.1
Saline water	9 LPM	Full	22.93	24.07	1.14	508.8
Saline water	26 LPM	Full	23.41	24.21	0.80	502.9
Saline water	9 LPM	Low	21.79	21.97	0.18	79.8

Saline water	26 LPM	Low	21.59	21.66	0.07	78.4

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